

Financially Constrained Households and Consumption Volatility in Open Economies^{*}

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Abstract

Emerging market economies often exhibit aggregate consumption that is more volatile than output, contrary to the consumption-smoothing prediction of standard macroeconomic models. This paper shows that heterogeneity in access to financial services can generate this excess volatility in a small open economy model driven by transitory productivity shocks alone. The mechanism is an unconstrained-amplification channel: as capital income is concentrated among fewer unconstrained households, their consumption unwinds more slowly after transitory shocks, raising its volatility relative to output. Procyclical firm entry provides a secondary amplification layer by strengthening the labor-income channel for hand-to-mouth households, but it is not the primary source of the result. Estimating the full model on macroeconomic data for 10 advanced, 17 emerging, and 11 low-income economies, we recover hand-to-mouth shares that rise with lower levels of development and align closely with independent microdata on financial access.

JEL classification: E13, E32, O11, O16, L11

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1. Introduction

Emerging market economies tend to have aggregate consumption that is more volatile than aggregate income, in contrast to the pattern observed in developed economies.¹ This empirical regularity presents a challenge to conventional macroeconomic frameworks, which predict consumption smoothing through financial markets when economies face transitory shocks to productivity. Previous research has attempted to reconcile this discrepancy by introducing additional sources of fluctuations that directly generate heightened consumption responsiveness.² In this paper, we pursue an alternative route motivated by the observation that access to financial services is far from universal in many emerging and low-income economies ([Honohan, 2008](#)). Specifically, we study how the presence of hand-to-mouth households alters the volatility of aggregate consumption in an otherwise standard small open economy real business cycle (SOE-RBC) framework driven by transitory productivity shocks alone.

The presence of hand-to-mouth households might appear to mechanically produce excess consumption volatility, as these households consume their entire income and therefore respond one-for-one to income shocks. However, this simple mechanism alone does not explain the cross-country pattern. Consider an economy parameterized so that consumption is already more volatile than output when all agents have full access to financial markets—the canonical advanced-economy case. Introducing hand-to-mouth households into this setup would reduce the aggregate consumption-to-output volatility ratio, since these households exhibit a relative volatility of one and pull the aggregate down. Conversely, starting from a parameterization where consumption is less volatile than income and adding hand-to-mouth households would raise the ratio toward, but not above, unity. Either way, the simple aggregation mechanism alone is inconsistent with the observed cross-country pattern, in which the consumption-to-output volatility ratio rises from 0.84 in developed economies to 1.04 in emerging and low-income economies. Resolving the puzzle therefore requires that the introduction of hand-to-mouth households alter not only the aggregation of consumption across types but also the consumption behavior of the unconstrained households who remain.

This paper makes two contributions. First, we identify and formalize a new mechanism

¹See [Uribe and Schmitt-Grohé \(2017\)](#) for detailed business cycle facts of developed and emerging market economies.

²To cite a few examples, [Neumeyer and Perri \(2005\)](#) include interest rate shocks, [Aguiar and Gopinath \(2007\)](#) include trend shocks to productivity, and [Garcia-Cicco, Pancrazi, and Uribe \(2010\)](#) include preference shocks.

through which household financial heterogeneity generates excess consumption volatility in response to transitory productivity shocks alone. The mechanism, which we call the unconstrained-amplification channel, operates not through a contemporaneous overreaction of consumption, the standard route in the existing literature, but through a slow and persistent unwinding of consumption in response to a transitory shock. We derive the channel analytically from the log-linearized Euler equation of unconstrained households. The key coefficient governing the speed at which unconstrained consumption returns to steady state is the consumption-gap ratio between hand-to-mouth and unconstrained households, which falls as the share of hand-to-mouth households rises and a fixed pool of capital income is concentrated in fewer hands. A smaller coefficient implies a weaker post-shock drag from hours growth on consumption, a fatter consumption impulse response, and a larger unconditional variance of unconstrained consumption relative to output. The channel is the dominant force behind excess consumption volatility in our simple model, and it explains why economies with more hand-to-mouth households also display more volatile consumption among the unconstrained.

Second, we estimate the model on macroeconomic time series spanning at least fifty years for a panel of 38 countries (10 advanced, 17 emerging, and 11 low-income), and recover a country-by-country posterior over the share of hand-to-mouth households. The estimated shares rise monotonically with lower levels of development: averaging 16% in advanced economies, 51% in emerging economies, and 59% in low-income economies. The shares also correlate strongly with the independent microdata measure of financial access constructed by [Honohan \(2008\)](#), with a cross-country correlation of 0.60. This external validation indicates that the macroeconomic time series of output, consumption, investment, and the trade balance, taken together, are informative about the share of financially constrained households in a manner consistent with direct survey-based measurement.

To identify the unconstrained-amplification channel cleanly, we first study a simple SOE-RBC model with household heterogeneity in financial access, following [Gali, López-Salido, and Vallés \(2004\)](#) and [Colciago \(2011\)](#).³ The production sector uses a CES aggregator of labor and capital with elasticity of substitution below one, generating empirically realistic labor-share dynamics. Beyond these features, the model retains standard ingredients: endogenous labor supply, investment in physical capital with convex adjustment costs, access to international financial markets, and a debt-elastic interest rate. The only source

³In the literature, financially constrained agents are also referred to as liquidity constrained, hand-to-mouth, rule-of-thumb, non-optimizing, or non-Ricardian households. Conversely, terms such as optimizing or Ricardian are used to describe unconstrained agents.

of exogenous disturbance is a transitory shock to total factor productivity. When calibrated to the typical emerging economy, the model generates aggregate consumption that is more volatile than aggregate output, consistent with empirical evidence and at realistic levels of shock persistence.⁴

We then conduct a formal variance decomposition of relative aggregate consumption volatility into three components: a hand-to-mouth term, an unconstrained-amplification term, and a covariance term. The decomposition reveals that the rise in relative consumption volatility as the share of hand-to-mouth households grows is not a mechanical consequence of aggregating in households with a fixed unit volatility ratio. Rather, it is driven by a sharp increase in the relative consumption volatility of unconstrained households themselves: the unconstrained-amplification term more than doubles as the share of hand-to-mouth households rises from 0.2 to 0.8. This finding is the empirical counterpart of the analytical mechanism derived from the Euler equation, and it constitutes the central contribution of the paper.

As a layered extension, we incorporate procyclical firm entry following [Jaimovich and Floetotto \(2008\)](#), which delivers endogenous markup variation and amplifies the cyclicity of the labor share.⁵ The augmented model preserves the unconstrained-amplification channel and adds a secondary amplification to the hand-to-mouth channel: by raising labor income relative to aggregate output during booms, firm dynamics push the consumption volatility of hand-to-mouth households above the volatility of aggregate output, which sustains $\sigma_c/\sigma_y > 1$ across a wider range of shock persistence values. The contribution of firm dynamics is quantitatively meaningful but secondary; in particular, firm dynamics on their own (without household heterogeneity) are insufficient to push the relative consumption volatility above one. The unconstrained-amplification channel remains the primary driver of the result.

To estimate the model on cross-country data, we extend the simple framework with firm dynamics along three dimensions following standard practice in the international macroeconomics literature: permanent productivity shocks, an interest-rate shock entering through the debt-elastic premium, and a working-capital constraint on firms. Beyond

⁴This is an important result since it is well known that increasing the persistence of the TFP shock induces agents to behave as if the change in income is close to permanent even in a standard SOE-RBC model, thus generating excess volatility of consumption (see [Uribe and Schmitt-Grohé, 2017](#) for a detailed explanation). The caveat of this strategy resides in the fact that high persistence of productivity generates counterfactually large autocorrelations of output.

⁵[Choi and Rios-Rull \(2009\)](#) and [Na \(2022\)](#) have shown that these labor share dynamics align with empirical observations.

the country-level estimates of the share of hand-to-mouth households described above, the estimation also treats the within- and across-sector elasticities of the [Jaimovich and Floetotto \(2008\)](#) firm-entry block as free parameters rather than calibrated values. This allows the macroeconomic time series to inform the labor-share-cyclicality margin, where direct cross-country measurement is currently constrained by limited data availability.

The model carries a transparent implication for stabilization policy, which we illustrate in the simple framework. Because aggregate shocks are transmitted through a distributional margin—a large share of households cannot smooth disposable income while unconstrained households absorb capital and external-financing risk—policies that stabilize aggregate absorption need not stabilize private consumption. We illustrate this with two stylized fiscal rules: a broad countercyclical public-absorption rule and a targeted transfer to hand-to-mouth households financed one-for-one by lower public absorption. The targeted transfer, which is countercyclical in its direct effect on constrained households’ disposable income, reduces both hand-to-mouth disposable-income volatility and aggregate relative consumption volatility, while the broad countercyclical rule does neither. The policy message is distributional rather than aggregative: stabilization tools that ignore the cyclical exposure of constrained households’ current income fail to address the source of the volatility we identify.

As a methodological robustness check, we also re-solve the simple model using a global solution method in addition to the perturbation method used for the main results, and find that the relative volatilities of consumption, investment, and the trade balance are very similar across solution methods. This addresses the concern raised by [De Groot, Durdu, and Mendoza \(2025\)](#) that perturbation methods combined with debt-elastic interest rates may bias consumption volatility upward.

Overall, our findings emphasize that excess consumption volatility in emerging and low-income economies arises from the interaction between transitory productivity shocks and household heterogeneity in financial access, operating through a slow unwinding of unconstrained consumption rather than through a contemporaneous overreaction. The interpretation reconciles, rather than displaces, alternative mechanisms based on permanent shocks or interest-rate shocks: those channels remain present and relevant in the full model that we estimate on cross-country data. What our framework adds is a distributional channel that delivers excess consumption volatility under transitory shocks alone, that is grounded in cross-country evidence on financial access, and that points to a corresponding distributional logic for stabilization policy.

Related literature. Our paper connects to several strands of the international macroeconomics literature on business cycles in emerging economies. A first strand documents and attempts to explain the consumption-to-output volatility ratio in open economies. [Uribe and Schmitt-Grohé \(2017\)](#) show that this ratio averages below one in developed economies but exceeds one in many emerging and low-income economies; we use the same business-cycle moments and combine them with the [Honohan \(2008\)](#) cross-country measure of household financial access to motivate the central model ingredient.

A second strand explores the role of permanent versus transitory productivity shocks in driving emerging-market business cycles. [Aguilar and Gopinath \(2007\)](#) argue that “the cycle is the trend” in emerging economies. [Garcia-Cicco et al. \(2010\)](#) push back by showing that, once a longer time series and a richer set of frictions are taken into account, stationary shocks remain the dominant driver of output fluctuations even in emerging economies. [Chang and Fernández \(2013\)](#) likewise find that transitory shocks are the dominant driver once the model is extended to include interest-rate and financial-friction shocks, and provide a horse-race comparison across competing specifications. [Fernández et al. \(2023\)](#) and [Singh \(2024\)](#) reach similar conclusions about the dominance of transitory shocks in different contexts. Motivated by this view, our paper focuses on whether transitory productivity shocks alone can generate excess consumption volatility once household heterogeneity is allowed for, complementing rather than replacing the channels emphasized in these papers.

A third strand introduces hand-to-mouth households into macroeconomic models to study how household heterogeneity in financial access shapes aggregate dynamics. [Gali et al. \(2004\)](#) and [Colciago \(2011\)](#) employ this approach in dynamic sticky-price settings to study monetary policy. More recently, [Bilbiie et al. \(2022\)](#) and [Bilbiie \(2025\)](#) develop two-agent New Keynesian (TANK) models that emphasize the role of hand-to-mouth households in amplifying business-cycle fluctuations through labor-income exposure. In the emerging economy context, [Bhattacharya and Patnaik \(2013\)](#) study the impact of hand-to-mouth households on consumption volatility in a closed emerging-market framework. Our paper differs from theirs in three respects: we work in a small open economy with external borrowing and lending, we identify a new channel (the unconstrained-amplification channel) that operates through the consumption-gap coefficient on the Euler equation, and our cross-country estimation of the share of hand-to-mouth households can be validated against independent microdata.

A fourth strand examines specific mechanisms that generate excess consumption

volatility under transitory shocks. [Neumeyer and Perri \(2005\)](#) and [Uribe and Yue \(2006\)](#) emphasize interest-rate shocks; [Michaud and Rother \(2018\)](#) study government-expenditure shocks; [Hong \(2020\)](#) combines transitory shocks with strong financial frictions and precautionary savings. The common feature of these channels is an amplified contemporaneous response of consumption relative to output at the onset of the shock. Our mechanism is distinctive in that it generates excess consumption volatility under transitory TFP shocks without requiring such a short-run overreaction. Instead, the operative channel is a slow unwinding of consumption driven by the Euler-equation coefficient on hours growth, which is absent in the alternative frameworks.

Finally, our work on procyclical firm entry and the labor share connects to the broader literature on time-varying factor shares over the business cycle. [Jaimovich and Floetotto \(2008\)](#) provide the production specification we use; [Choi and Rios-Rull \(2009\)](#), [Rios-Rull and Santaella-Llopis \(2010\)](#), [Na \(2022\)](#), and [Karabarbounis and Neiman \(2014\)](#) document the cyclical behavior of the labor share across countries. The plant-level evidence of [Ates and Saffie \(2021\)](#) confirms that firm entry falls sharply during sudden stops in emerging economies, an observation consistent with the firm-dynamics block we use as an amplification layer.

Outline. The rest of the paper is organized as follows. Section 2 presents the empirical motivation by analyzing business cycle statistics and household financial access across a broad set of advanced, emerging, and low-income economies. Section 3 introduces the simple illustrative model and demonstrates how household heterogeneity in financial access generates excess consumption volatility under transitory TFP shocks alone; it also formalizes the unconstrained-amplification channel via the Euler equation, presents a variance decomposition of relative aggregate consumption volatility, and layers in the firm-entry block as an amplification mechanism. Section 4 extends the simple model with additional shocks, estimates the resulting full model on a panel of 38 countries, and reports the country-level estimates of the share of hand-to-mouth households. Section 5 discusses the policy implications. Section 6 concludes.

2. Empirical motivation

We document two empirical facts that motivate the model. First, aggregate consumption is substantially more volatile relative to output in emerging and low-income economies

than in developed economies. Second, this excess consumption volatility is larger in countries where fewer households have access to formal financial services. Appendix A.1 reports details on the data and the construction of moments.

The business-cycle statistics are computed from annual national accounts data for 38 countries with long and continuous series on output, consumption, investment, and the trade balance. Table 1 summarizes the main moments. Output is more volatile in emerging and low-income economies, with a standard deviation of 3.3%, compared with 1.9% in developed economies. The key moment for the paper is the relative volatility of consumption, which rises from 0.84 in developed economies to 1.04 in emerging and low-income economies.

Table 1: Business cycle statistics for countries with different levels of development

Statistics at the yearly frequency		Country development	
		Developed	Emerging
Standard deviation of output	$SD(y)$	1.9	3.3
Autocorrelation of output	$AC(y)$	0.55	0.48
Relative standard deviation:			
Consumption	$SD(c) / SD(y)$	0.84	1.04
Investment	$SD(i) / SD(y)$	2.65	3.20
Trade balance	$SD(TB/Y) / SD(y)$	0.52	0.72
Correlation with output:			
Consumption	$C(y, c)$	0.73	0.74
Investment	$C(y, i)$	0.84	0.67
Trade balance	$C(y, TB/Y)$	-0.31	-0.23
Number of countries	N	10	28

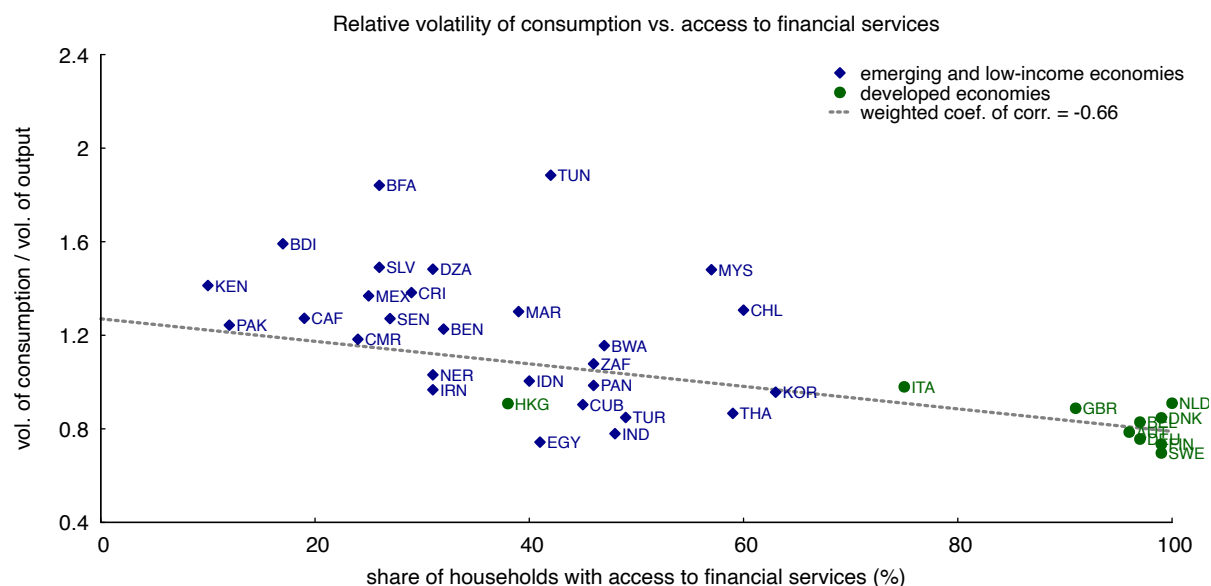
Source: *World Development Indicators* from the World Bank.

Notes: This table computes business cycle statistics for all countries in the sample. All statistics apply to the cyclical component of the corresponding variable extracted by applying an HP-filter. The numbers in the table show weighted averages across countries in each group, using each country's share of world GDP as weights. The column labeled "Emerging" pools the 17 emerging and 11 low-income economies in the sample.

Although consumption volatility is the central object of the paper, the remaining moments in Table 1 discipline the model. Investment and the trade balance are also more volatile relative to output in emerging and low-income economies, while their correlations with output are weaker. The model therefore has to match the consumption-volatility fact without generating implausible investment or external-adjustment dynamics.

Figure 1 links this macroeconomic pattern to household financial access. We use the estimates of [Honohan \(2008\)](#), which proxy for the share of households able to use formal financial intermediaries. In the model, the complement of this measure maps naturally into the fraction of hand-to-mouth households: a country with lower measured access is interpreted as having more households that cannot smooth income shocks through saving or borrowing.

Figure 1: Relationship between the relative volatility of consumption and access to financial services across countries



Source: *World Development Indicators* from the World Bank and [Honohan \(2008\)](#).

Notes: The scatter plot shows the share of households with access to financial services, as measured by [Honohan \(2008\)](#), against the ratio of the volatility of consumption to the volatility of output, measured from national accounts data. The trend line is derived from a regression of the two variables, weighted by each country's share of world GDP.

The figure shows a clear negative relationship. Economies with broad access to financial services tend to have consumption volatility below output volatility, while countries with limited access are more likely to display excess consumption volatility. The relationship is visible both in the fitted line and in the location of the country groups: developed economies cluster toward high access and lower relative consumption volatility, whereas many emerging and low-income economies lie further to the left and higher in the figure. The correlation is -0.63 without weights and -0.66 when observations are weighted by GDP. We do not interpret this bivariate relationship as causal evidence; rather, it motivates the central model ingredient in Section 3: cross-country variation in the fraction of financially

constrained households.

3. A simple illustrative model

This section introduces a parsimonious small open economy model that clarifies how financial constraints can generate excess consumption volatility in response to transitory productivity shocks. This focus is motivated by evidence that such shocks are important for business cycles in emerging and low-income economies (Chang and Fernández, 2013; Garcia-Cicco et al., 2010; Fernández et al., 2023). In standard SOE-RBC models with full access to international credit markets, households smooth transitory income fluctuations and consumption is typically less volatile than output. Uribe and Schmitt-Grohé (2017), for example, show that transitory shocks can generate excess consumption volatility only under very high shock persistence.

We keep the environment close to the standard international RBC framework and add two ingredients. First, households differ in their access to financial markets: a fraction λ is hand-to-mouth, while the remaining households can save, borrow, and accumulate capital. Second, production uses a CES technology, so factor shares can move over the cycle. These two ingredients make the model useful for isolating the distributional channel emphasized in the paper: shocks affect labor and capital income differently, and limited financial access determines how much of that income variation is transmitted directly into aggregate consumption. Appendix A.4 reports the full baseline equilibrium conditions and steady state.

Households. The economy consists of two types of households: hand-to-mouth households, which make up a fraction λ of the population, and unconstrained households, comprising the remaining fraction $1 - \lambda$. Both household types have GHH preferences, as in Greenwood, Hercowitz, and Huffman (1988),

$$u(c_t^i, h_t^i) = (c_t^i - (h_t^i)^\nu / \nu)^{1-\gamma} / (1 - \gamma), \quad i \in \{htm, unc\}. \quad (1)$$

The GHH specification eliminates wealth effects on labor supply. As a result, both household types satisfy the same static labor supply condition, $w_t = (h_t^i)^{\nu-1}$, and supply the same hours in equilibrium. We use this preference specification as an isolating assumption rather than as an additional amplification mechanism: differences across household

types then arise from asset-market access and income composition, not from different labor-supply wealth effects.

Hand-to-mouth households cannot save, borrow, or accumulate capital. They consume current labor income,

$$c_t^{htm} = w_t h_t^{htm} \quad (2)$$

where w_t is the wage. Unconstrained households supply labor, rent capital, invest subject to convex adjustment costs, and trade one-period foreign debt. Their intertemporal optimality conditions are standard and are listed in Appendix A.4. The key economic difference between the two household types is therefore not labor supply, but access to assets and capital income.

Production. Output is produced by a competitive firm that hires labor, $\tilde{l}_t$, rents capital, $\tilde{k}_t$, and combines them using a CES technology:

$$\tilde{y}_t = a_t \cdot \left(\alpha \tilde{k}_t^{\frac{\zeta-1}{\zeta}} + (1-\alpha) \tilde{l}_t^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (3)$$

where a_t is a common productivity shock.⁶ The CES structure allows the labor share to move over the cycle. In particular, profit maximization implies the labor share

$$sh_t^{\tilde{l}} = \frac{(1-\alpha)}{(1-\alpha) + \alpha \left(\frac{\tilde{k}_t}{\tilde{l}_t} \right)^{\frac{\zeta-1}{\zeta}}} \quad (4)$$

which is not mechanically constant when $\zeta \neq 1$. For $\zeta < 1$, the predetermined capital stock makes the labor share initially fall after a positive productivity shock, but the labor share can recover and overshoot as capital and labor adjust. This cyclical movement in factor shares matters because hand-to-mouth households consume labor income contemporaneously.

⁶As pointed out by [Choi and Rios-Rull \(2009\)](#), a constant elasticity of substitution production function generates models with more realistic business cycle dynamics. The Cobb-Douglas case emerges when $\zeta \rightarrow 1$.

Aggregation and equilibrium. Aggregate consumption is the weighted average of each household type's consumption:

$$c_t = \lambda c_t^{htm} + (1 - \lambda) c_t^{unc}. \quad (5)$$

This identity is the central object in the mechanism analysis below. When λ is high, a larger share of aggregate consumption comes from households whose consumption tracks current labor income. At the same time, the consumption of unconstrained households adjusts to capital accumulation and foreign borrowing decisions. Market clearing equates aggregate capital and labor supply with firm demand, and the trade balance is defined as net external asset accumulation, $tb_t = d_t - d_{t+1}/(1 + r_t)$.

Shocks and international credit. The only exogenous source of fluctuations is a transitory productivity shock,

$$\log a_{t+1} = \rho_a \log a_t + \sigma_a \epsilon_{at+1}, \quad \epsilon_{at+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

where ρ_a and σ_a denote persistence and conditional volatility. We close the small open economy using the standard debt-elastic interest-rate device:

$$r_t = r^* + \psi \cdot \left(e^{\frac{d_{t+1} - \bar{d}}{y}} - 1 \right)$$

where $r^* \equiv 1/\beta - 1$ is the world interest rate and $\psi > 0$ governs the response of borrowing costs to deviations of debt from its steady state. For the global-solution robustness exercise, we instead set $r_t = r^*$ and impose $\beta(1 + r^*) < 1$ for stationarity.

3.1. Model solution

We follow [Uribe and Schmitt-Grohé \(2017\)](#) and solve the simple model using the perturbation method. As a robustness check, we also apply a global solution method to the setup and compare the resulting moments across solution methods.

First, we parameterize the model to highlight key business cycle statistics generated by a typical emerging economy. While our primary focus is on the volatility of aggregate consumption relative to the volatility of output, $\sigma(c)/\sigma(y)$, we also report second moments for output, investment, and the trade balance, following standard practice in the

international macroeconomics literature. The main model parameters remain the same across the two solution methods and are chosen using standard values from previous studies and typical empirical targets. A full estimation of the model is deferred to Section 4.

Table 2 presents the parameter choices for this exercise. The annual real interest rate is fixed at $r^* = 7\%$, reflecting the U.S. risk-free rate of 4% plus a 3% risk premium. This interest rate maps into a discount factor of $\beta = 0.9346$ when we use the perturbation method. Under the global solution method, stationarity is instead imposed by assuming $\beta(1 + r^*) < 1$.⁷ We therefore assume a discount factor of $\beta = 0.9328$ when we solve the model using the global solution method. This value is in line with the one used in [Uribe and Schmitt-Grohé \(2017\)](#) when they impose stationarity in a similar RBC model using global methods. The chosen value of the discount factor generates $\beta(1 + r^*) = 0.9981$, which remains very close to unity while also satisfying the stationarity condition.

Table 2: Parameterization of the simple model

Parameter		Value	Target
International interest rate	r^*	0.07	Standard
Discount factor (perturbation)	β	0.9346	$\beta(1 + r^*) = 1$
Discount factor (global solution)	β	0.9328	$\beta(1 + r^*) = 0.9981$
Consumption intertemporal elasticity	γ	2	Mendoza (1991)
Labor supply elasticity	ν	1.4	Mendoza (1991)
Capital depreciation rate	δ	0.15	Standard
Production function capital share	α	0.475	Labor share of 65%
Elasticity of substitution capital/labor	ζ	0.6	Choi and Rios-Rull (2009)
Capital convex adjustment cost	ϕ	0.001	Rel. vol. of investment of 3.2
Interest rate debt elasticity	ψ	0.004	Rel. vol. of trade balance of 0.7
External debt to output	$\bar{d}/\bar{y}$	0.7	Lane and Milesi-Ferretti (2022)
Standard deviation of output shock	σ_a	0.0175	GDP volatility of 3.3%
Persistence of output shock	ρ_a	0.17	GDP persistence of 0.48
Share of constrained households	λ	0.65	1 – average share from Fig 1

The remaining parameters are the same across the two solution methods. We follow

⁷The detailed description of the global solution method and the assumptions is presented in Section A.4.4.

Mendoza (1991) in setting the coefficient governing the intertemporal elasticity of substitution, γ , to 2, and the parameter controlling the Frisch elasticity of labor supply, ν , to 1.4. These values correspond to an intertemporal elasticity of 0.5 and a Frisch elasticity of 2.5, both of which are standard in the literature.

We also adopt the conventional value of 15% for the depreciation rate, δ . For the CES production function, the capital share parameter, α , is set to 0.475 to ensure a steady-state labor share of 65%. Following Klump et al. (2007), Chirinko (2008), and Choi and Rios-Rull (2009), we set the elasticity of substitution between capital and labor, ζ , to 0.6. The convex capital adjustment cost parameter is chosen as $\phi = 0.001$, targeting an investment volatility of 3.2 times that of output. The debt elasticity of the interest rate ($\psi = 0.004$) is chosen to reflect a typical emerging market economy, with a trade balance-to-output volatility of 70% of the output volatility.

The steady-state debt-to-GDP ratio, $\bar{d}/\bar{y}$, is set to 0.7, corresponding to the average ratio of gross external debt liabilities (defined as the sum of portfolio debt liabilities and other investment liabilities) to GDP across 98 emerging and low-income economies over the recent 40 years (1983-2022), computed from the dataset of Lane and Milesi-Ferretti (2018) and Lane and Milesi-Ferretti (2022). Finally, the shock parameters $\sigma_a = 0.0175$ and $\rho_a = 0.17$ are selected to replicate an output volatility of 3.3% with a persistence of 50%, consistent with the results in Table 1. The fraction of hand-to-mouth agents, λ , is set to 0.65, consistent with Figure 1, which shows that the average share of households with access to financial services in emerging economies is about 35%.

3.1.1. Model performance: targeted and non-targeted moments

Using the parameter values from Table 2, we compute the theoretical moments implied by the policy functions at an annual frequency. All moments are expressed in terms of the log-deviation of each variable from its steady state, except the trade-balance-to-output ratio, which is expressed as a deviation from its steady state. Table 3 presents a selection of these moments, highlighting that the model—driven solely by transitory productivity shocks—can closely replicate key business cycle statistics observed in emerging and low-income economies. For comparison, the second column of Table 3 reports the empirical moments from Section 2 for a typical emerging and low-income economy.

Among the targeted moments, the baseline parameterization matches all four moments from the data. The standard deviation (3.3 in the model and the data) and autocorrelation (0.53 in the model and 0.48 in the data) of output are a close match, validating the choice

Table 3: Business cycle moments: data vs model

Moments		Data	Model	
			Baseline $\lambda = 0.65$	No heterogeneity $\lambda = 0$
Targeted moments				
Standard deviation of output	$\sigma(y)$	3.3	3.3	3.3
Autocorrelation of output	$\rho(y)$	0.48	0.53	0.54
Relative standard deviation:				
Investment	$\sigma(i)/\sigma(y)$	3.20	3.24	3.24
Trade balance	$\sigma(tby)/\sigma(y)$	0.72	0.72	0.73
Non-targeted moments				
Relative standard deviation:				
Consumption	$\sigma(c)/\sigma(y)$	1.04	1.05	0.96
Correlation with output:				
Consumption	$\rho(y, c)$	0.74	0.85	0.80
Investment	$\rho(y, i)$	0.67	0.50	0.50
Trade balance	$\rho(y, tby)$	-0.23	-0.09	0.02

Notes: The column 'Baseline' shows the moments generated by the baseline model based on the parameterization shown in Table 2, whereas the column 'No heterogeneity' keeps the same parameters except for the share of constrained households, which is set to zero, $\lambda = 0$.

of standard deviation and persistence of the productivity shock. A value of 3.24 for the relative volatility of investment in the baseline model also matches the value of 3.20 from the data, confirming the choice of capital adjustment cost parameter. Last, the relative volatility of the trade-balance-to-output ratio shows a close match between the model (0.72) and the data (0.72), supporting the choice of debt elasticity parameter.

Notably, among the non-targeted moments, the baseline model performs exceptionally well in capturing the excess volatility of consumption, which is the central focus of our analysis (1.04 in the data vs 1.05 in the model). Crucially, the model is capable of generating excess consumption volatility even when the persistence of output remains at a realistically moderate level. Beyond consumption volatility, the model does well at reproducing other defining features of emerging and low-income economies. For example, the baseline model displays highly procyclical consumption (0.74 in the data vs 0.85 in the model), moderately procyclical investment (0.67 in the data vs 0.50 in the model), and slightly

countercyclical trade-balance-to-output ratio (-0.23 in the data vs -0.09 in the model). These features show a reasonable match with the data.

3.1.2. Model performance: perturbation vs global solution method

As a robustness exercise, we compare the results of the baseline model when we use an alternative solution method, the global method. The results are presented in Table 9 in Appendix A.2. The stationarity device differs across the two exercises. The perturbation solution uses the debt-elastic interest-rate specification, while the global solution sets $r_t = r^*$ and uses impatience, $\beta(1 + r^*) < 1$, to induce stationarity. Apart from this change in the stationarity device and the associated small adjustment in the discount factor, from 0.9346 to 0.9328, the remaining parameters are kept fixed.

Most targeted volatility moments remain close across solution methods. Crucially, the central non-targeted moment, the relative volatility of consumption, is essentially unchanged: 1.05 under perturbation and 1.06 under the global method. The relative volatilities of investment and the trade balance are also close, at 3.24 versus 3.15 and 0.72 versus 0.74, respectively. Output volatility and output persistence remain similar as well. Some non-targeted correlations differ more. In particular, the investment-output correlation falls from 0.50 under perturbation to 0.28 under the global method, and the trade-balance-output correlation moves from -0.09 to 0.13.⁸

Our results are in line with Uribe and Schmitt-Grohé (2017) and De Groot et al. (2025), who show that many cyclical moments can be similar across solution methods while some dynamics differ. We therefore interpret this exercise as evidence that the paper’s relative-consumption-volatility mechanism is not materially driven by the local perturbation solution. We use the parameterization of Table 2 and the perturbation method as the benchmark for exploring the mechanisms driving excess consumption volatility.

⁸As a comparison, when Uribe and Schmitt-Grohé (2017) apply the two solution methods for a similar model on Canadian data, they obtain 0.83, 2.44, and 0.35 as the relative volatility of consumption, investment, and trade-balance-to-output ratio using global methods. The corresponding perturbation values are 0.88, 2.94, and 0.58. The standard deviation and persistence of output are 3.56 and 0.68 under global methods, compared with 3.08 and 0.62 under perturbation. The correlations of output with consumption, investment, and trade-balance-to-output ratio are 0.98, 0.69, and -0.04 using global methods and 0.84, 0.67, and -0.04 using perturbation. De Groot et al. (2025) show an even closer match, but their basic model features an endowment economy.

3.1.3. Model performance: baseline vs no heterogeneity model

We next assess the role of household heterogeneity in generating excess consumption volatility. We start by comparing business-cycle moments with and without household heterogeneity. We then vary the degree of heterogeneity and examine the mechanism that drives excess consumption volatility.

The last column of Table 3 reports moments for a version of the model without household heterogeneity, obtained by setting $\lambda = 0$ while keeping all other parameters unchanged. In this case, the volatility of consumption no longer exceeds that of output, with the ratio declining from 1.05 in the baseline model to 0.96 in the no-heterogeneity model. This result is consistent with standard international RBC models that do not incorporate additional sources of shocks. Furthermore, the trade balance-to-output ratio becomes slightly procyclical, with a correlation between output and the trade-balance-to-output ratio of 0.02, compared with the countercyclicity of the trade balance shown in the baseline model. These findings highlight that the RBC model without heterogeneity fails to capture the macroeconomic dynamics that are characteristic of emerging market economies, especially in the absence of other sources of shocks. Household heterogeneity is therefore crucial when the economy is driven by transitory shocks to output.

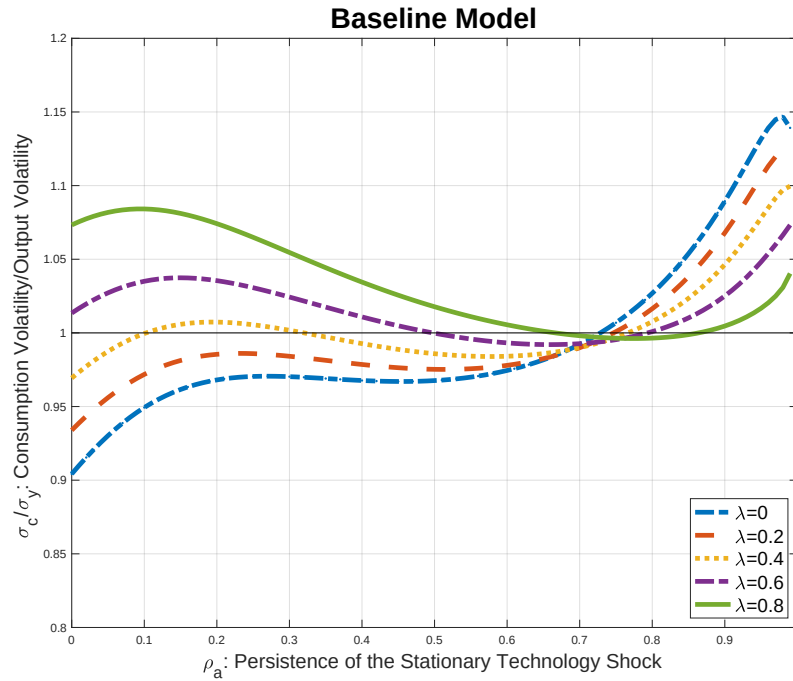
3.2. Consumption volatility: the role of heterogeneity

Because the baseline model with household heterogeneity matches the consumption-volatility fact better than its homogeneous-household counterpart, we now examine how heterogeneity contributes to excess consumption volatility. We proceed in two stages. First, we examine how the relative volatility of consumption responds to variations in the degree of household heterogeneity, governed by the parameter λ , and to the persistence of output realizations, governed by ρ_a . Second, we conduct a formal decomposition of relative consumption volatility to identify the mechanisms through which household heterogeneity amplifies consumption volatility beyond the benchmark predicted by the homogeneous specification.

A question of particular importance is whether the relative volatility of consumption can exceed unity under empirically plausible levels of persistence for transitory productivity shocks. This question is motivated by the findings of [Uribe and Schmitt-Grohé \(2017\)](#), who demonstrate that a standard real business cycle (RBC) model driven exclusively by transitory shocks is capable of generating excess consumption volatility only when shock

persistence is implausibly high, exceeding a threshold of 0.9. To benchmark our baseline model against their results, we consider an augmented version of their parsimonious RBC framework that incorporates household heterogeneity and a CES production function. Figure 2 presents the results of this exercise, illustrating how excess consumption volatility responds to variations in the share of hand-to-mouth households (λ) and the degree of persistence in the productivity shock process (ρ_a).

Figure 2: Ratio of consumption volatility and output volatility and the relationship with persistence of transitory/stationary technology shock in the model



Notes: This figure shows how changes in the persistence of the technology shock given by ρ_a (on the horizontal axis) affect the model-generated ratio of the volatility of consumption to the volatility of output given by $\mathbb{SD}(c)/\mathbb{SD}(y)$. The results are based on the parameterization summarized in Table 2 along with different combinations of λ and ρ_a .

The figure shows that the baseline model generates excess consumption volatility across a wide range of parameter configurations. When the share of financially constrained agents is 60% or 80% (indicated by the $\lambda = 0.6$ and $\lambda = 0.8$ lines), the baseline model generates consumption volatility that exceeds output volatility even for low persistence of transitory productivity shocks. Even when the share of financially constrained agents is 40%, relative consumption volatility exceeds unity in the baseline model for reasonable persistence levels (ρ_a higher than 0.1). This result underscores a central contribution of

the paper: whereas conventional models require transitory shock persistence in excess of 0.9 to generate excess consumption volatility, our model achieves this outcome under substantially more plausible parametric conditions.

To clarify the mechanisms underlying excess consumption volatility, we use the aggregate consumption identity

$$c = \lambda c^{htm} + (1 - \lambda) c^{unc},$$

and decompose the relative variance of aggregate consumption into three distinct components.

$$\left(\frac{\sigma_c}{\sigma_y}\right)^2 = \underbrace{\left(\frac{\lambda c_{ss}^{htm}}{c_{ss}}\right)^2 \left(\frac{\sigma_{c^{htm}}}{\sigma_y}\right)^2}_{\text{HTM volatility}} + \underbrace{\left(\frac{(1 - \lambda) c_{ss}^{unc}}{c_{ss}}\right)^2 \left(\frac{\sigma_{c^{unc}}}{\sigma_y}\right)^2}_{\text{UNC volatility}} + \underbrace{2\lambda(1 - \lambda) \left(\frac{c_{ss}^{htm}}{c_{ss}}\right) \left(\frac{c_{ss}^{unc}}{c_{ss}}\right) \frac{\sigma_{c^{htm}} \sigma_{c^{unc}}}{\sigma_y^2} \rho_{htm,unc}}_{\text{Covariance}}$$

The first term captures the contribution of hand-to-mouth household consumption volatility, weighted by their steady-state consumption share. The second term captures the analogous contribution from unconstrained households. The third term represents the covariance between the two, scaled by their respective steady-state shares and governed by the cross-correlation $\rho_{htm,unc}$. Panel A of Table 4 presents the decomposition across varying degrees of household heterogeneity, and we discuss the behavior of each term in turn.

First term: the hand-to-mouth channel. Since constrained households consume their labor income each period, their consumption tracks labor income almost one-for-one, so $\sigma_{c^{htm}}/\sigma_y$ is close to, but slightly below, unity. The mild departure arises from the CES production structure and relatively small persistence of the productivity shock. The steady-state consumption share c_{ss}^{htm}/c_{ss} is also stable across values of λ , since hand-to-mouth consumption in steady state depends only on the wage and hours worked, both of which are independent of λ . As a result, the contribution of Term 1 is driven almost entirely by a mechanical scaling effect: as λ increases fourfold from 0.2 to 0.8, Term 1 rises approximately sixteen-fold, consistent with its quadratic dependence on λ . In isolation, one would expect a symmetric and offsetting decline in Term 2 to neutralize this amplification. As we discuss next, however, this intuition is overturned by the behavior of unconstrained households.

Table 4: Decomposition of relative variance of aggregate consumption

λ	HTM volatility			UNC volatility			Covariance		Relative variance
	$\frac{c_{ss}^{htm}}{c_{ss}}$	$\frac{\sigma_{c^{htm}}}{\sigma_y}$	Term 1	$\frac{c_{ss}^{unc}}{c_{ss}}$	$\frac{\sigma_{c^{unc}}}{\sigma_y}$	Term 2	$\rho_{htm,unc}$	Term 3	$(\sigma_c/\sigma_y)^2$
<i>Panel A: baseline model, $\rho_a = 0.17$</i>									
0.2	0.9088	0.9065	0.0271	1.0228	1.0381	0.7215	0.7812	0.2187	0.9673
0.4	0.9088	0.9063	0.1086	1.0608	1.1687	0.5533	0.7184	0.3521	1.0139
0.6	0.9088	0.9063	0.2443	1.1368	1.4383	0.4277	0.6242	0.4035	1.0755
0.8	0.9088	0.9067	0.4346	1.3647	2.1659	0.3495	0.4870	0.3796	1.1636
<i>Panel B: baseline model with firm dynamics, $\rho_a = 0.17$</i>									
0.2	0.9094	0.9660	0.0309	1.0227	1.0165	0.6916	0.7902	0.2309	0.9534
0.4	0.9094	0.9658	0.1234	1.0604	1.1289	0.5160	0.7246	0.3657	1.0051
0.6	0.9094	0.9658	0.2777	1.1360	1.3663	0.3854	0.6217	0.4067	1.0698
0.8	0.9094	0.9659	0.4937	1.3626	2.0232	0.3040	0.4631	0.3589	1.1566
<i>Panel C: baseline model, $\rho_a = 0.70$</i>									
0.2	0.9088	1.0236	0.0346	1.0228	1.0092	0.6818	0.8576	0.2635	0.9800
0.4	0.9088	1.0231	0.1383	1.0608	1.0403	0.4384	0.8192	0.4034	0.9802
0.6	0.9088	1.0225	0.3109	1.1368	1.1066	0.2532	0.7509	0.4213	0.9854
0.8	0.9088	1.0219	0.5521	1.3647	1.2963	0.1252	0.6066	0.3189	0.9961
<i>Panel D: baseline model with firm dynamics, $\rho_a = 0.70$</i>									
0.2	0.9094	1.0889	0.0392	1.0227	1.0654	0.7598	0.8552	0.2952	1.0942
0.4	0.9094	1.0882	0.1567	1.0604	1.0989	0.4888	0.8161	0.4517	1.0972
0.6	0.9094	1.0876	0.3521	1.1360	1.1700	0.2826	0.7466	0.4711	1.1059
0.8	0.9094	1.0870	0.6253	1.3626	1.3734	0.1401	0.6001	0.3552	1.1205

Notes: The table decomposes the relative variance of aggregate consumption, $(\sigma_c/\sigma_y)^2$, into three terms. Term 1 captures the contribution of hand-to-mouth household consumption volatility, Term 2 captures the contribution of unconstrained household consumption volatility, and Term 3 captures the covariance between the two. Panels A and C report results for the baseline model at $\rho_a = 0.17$ and $\rho_a = 0.70$ respectively. Panels B and D report the corresponding results for the augmented model with firm dynamics. All remaining parameters follow Table 2, with $\alpha = 0.11$ and $\sigma_a = 0.0125$ in Panels B and D.

Second term: the unconstrained amplification. The expectation that Term 2 would decline in proportion to $(1 - \lambda)^2$, fully offsetting the rise in Term 1, is not borne out. Instead, as λ increases from 0.2 to 0.8, two forces emerge within Term 2. First, the steady-state consumption share of unconstrained households, c_{ss}^{unc}/c_{ss} , rises from approximately 1.02 to 1.36, reflecting the concentration of capital income among a shrinking population. Second, and more strikingly, the relative consumption volatility $\sigma_{c^{unc}}/\sigma_y$ more than doubles, rising from 1.04 to 2.17. These two forces together ensure that Term 2 declines only by half, from

0.72 to 0.35, rather than the sixteen-fold reduction that mechanical scaling alone would predict. It is this sharp amplification in unconstrained household consumption volatility that constitutes the central and novel mechanism of the paper. We turn to its explanation in the following section.

Third term: the covariance. The covariance term corroborates the dominance of the unconstrained amplification. Although the factor $\lambda(1 - \lambda)$ peaks near $\lambda = 0.5$ and the cross-correlation $\rho_{htm,unc}$ declines monotonically with λ , Term 3 nevertheless increases. This is attributable to the concurrent rise in both c_{ss}^{unc}/c_{ss} and $\sigma_{c^{unc}}/\sigma_y$, which more than offsets these damping forces. All three terms therefore point to the same underlying source: the increasingly volatile consumption of unconstrained households as the economy becomes more unequal.

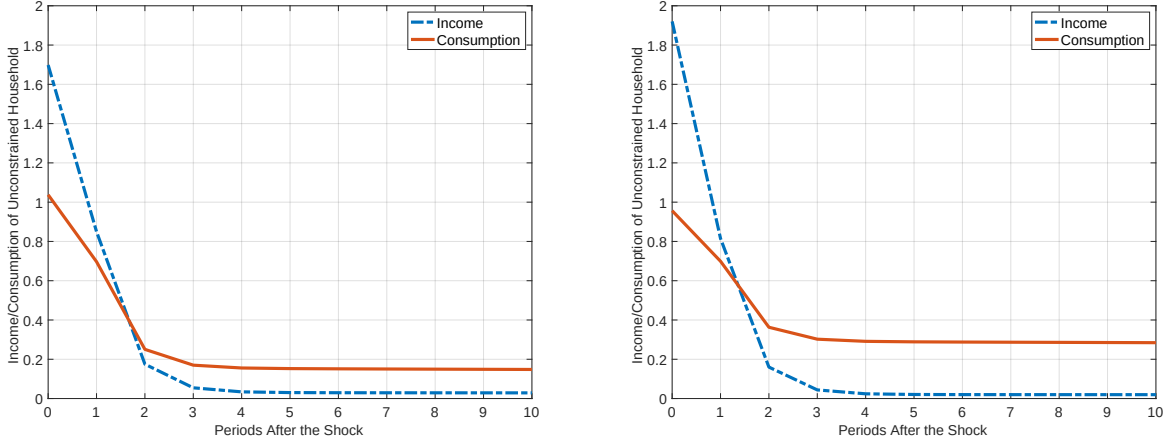
3.3. The unconstrained amplification

The decomposition exercise reveals a fundamental departure from standard intuition: the rise in aggregate consumption volatility is not merely a mechanical consequence of a larger share of households unable to smooth consumption, but rather stems from an amplification in the relative consumption volatility of those who can. A central question therefore arises: why does a larger share of hand-to-mouth agents so strongly change the consumption behavior of unconstrained households, making their consumption more than twice as volatile as output? The remainder of this section answers this question.

We present impulse response functions (IRFs) of key variables from the baseline model to visually illustrate the mechanisms driving our results. For the parameterization from Table 2, a positive TFP shock generates broadly similar dynamics for aggregate output regardless of the share of hand-to-mouth households, λ , as shown in Figure 3. The response of unconstrained household consumption, however, differs markedly across economies. In high- λ economies, unconstrained households shift consumption toward the future rather than the present: while the on-impact response of consumption is smaller relative to output, consumption remains elevated well beyond the peak of the shock, and its subsequent slow unwinding inflates the unconditional variance of unconstrained consumption relative to output.⁹ Table 4 confirms this, documenting a near-doubling of the relative volatility of consumption for unconstrained households relative to aggregate income as λ increases.

⁹De Groot et al. (2025) show that the medium- and long-run response of consumption is essentially unchanged when the model is solved globally rather than by perturbation.

Figure 3: Impulse response to 1% transitory shock in productivity



Notes: Both panels show impulse response functions of income and consumption of unconstrained households. The left panel uses $\lambda = 0.2$ while the right panel uses $\lambda = 0.8$. All variables are shown as percentage deviations from steady-state values. The results are from the baseline model based on the parameterization summarized in Table 2.

The intuition for this persistence is rooted in household wealth composition. As λ rises, unconstrained households are fewer but absorb all capital income, making each unconstrained household wealthier and tilting their income toward capital rather than labor. A household relying primarily on labor income must track hours closely: as the productivity boom fades, hours and wages fall, leaving little room to sustain elevated consumption. A wealthier unconstrained household, by contrast, has a greater share of income coming from capital. Its consumption therefore need not fall in lockstep with hours; instead, it drifts down gradually. In high- λ economies, it is this slow unwinding—not a larger peak response—that drives excess consumption volatility.

This mechanism represents a departure from the existing literature on excess consumption volatility, where the dominant channel is typically an on-impact overreaction of consumption relative to output (Uribe and Schmitt-Grohé, 2017). Here, the peak consumption response is actually smaller than the peak output response; it is the slow, persistent unwinding of consumption, not an initial spike, that inflates the unconditional variance.

To formalize this channel, we log-linearize the Euler equation of unconstrained households under GHH preferences:

$$\hat{c}_{t+1}^{unc} - \hat{c}_t^{unc} = \frac{h_{ss}^\nu}{c_{ss}^{unc}} \cdot (\hat{h}_{t+1} - \hat{h}_t) + \frac{c_{ss}^{unc} - h_{ss}^\nu/\nu}{\gamma c_{ss}^{unc}} \cdot \hat{r}_t$$

After a transitory TFP shock, hours rise on impact but quickly revert, so hours growth is negative, $\hat{h}_{t+1} - \hat{h}_t < 0$, for all periods after the initial impact of the shock. Simultaneously, households accumulate savings (because the productivity boom is transitory) and reduce debt below target, pushing the interest rate below steady state ($\hat{r}_t < 0$). Both terms therefore exert downward pressure on consumption growth, pulling the consumption of unconstrained agents back toward steady state. Since the debt-elasticity parameter ψ is small in the EDEIR specification, the interest rate term is quantitatively small and the hours term dominates.

The key coefficient linking hours to consumption growth is h_{ss}^v/c_{ss}^{unc} —the ratio of labor disutility to total household resources. As λ rises, c_{ss}^{unc} increases because unconstrained households absorb a fixed pool of capital income across a shrinking population, while aggregate hours h_{ss} are essentially unaffected by λ as labor markets clear at the economy-wide level. Consequently, h_{ss}^v/c_{ss}^{unc} falls with λ .

Writing the key coefficient h_{ss}^v/c_{ss}^{unc} as $w_{ss}h_{ss}/c_{ss}^{unc} \equiv c_{ss}^{htm}/c_{ss}^{unc}$ clarifies its economic content. As the fraction of credit-constrained households increases, the consumption gap between constrained and unconstrained households widens (equivalently, $c_{ss}^{htm}/c_{ss}^{unc}$ falls). Since unconstrained households' consumption is financed increasingly by capital income, the impact of hours growth on consumption growth decreases.

A smaller coefficient means that declining hours exert a weaker drag on unconstrained consumption growth after the shock. Summing the Euler equation across periods, the cumulative return of c^{unc} to steady state is slower when λ is large, generating a fatter and more persistent impulse response. This persistence directly inflates $\sigma_{c^{unc}}/\sigma_y$, explaining why excess consumption volatility among unconstrained households rises monotonically with the degree of financial exclusion.

3.4. Amplifying the hand-to-mouth channel: procyclical firm dynamics

The decomposition in Table 4 reveals that the hand-to-mouth channel, while growing mechanically with λ , is quantitatively constrained by the fact that the consumption volatility of constrained households remains slightly below aggregate output volatility, $\sigma_{c^{htm}}/\sigma_y < 1$. This occurs because constrained households consume their labor income each period, and labor income is less volatile than aggregate output under a CES production structure with less persistent shocks. In other words, the hand-to-mouth channel cannot on its own push aggregate consumption volatility substantially above output volatility; its amplification potential is bounded by how volatile labor income and the labor share are

relative to aggregate output.

This observation motivates a natural extension: if one can make labor income more volatile than aggregate output, the hand-to-mouth channel becomes a more powerful driver of excess consumption volatility. We show that procyclical firm dynamics, of the kind studied by [Jaimovich and Floetotto \(2008\)](#), provide exactly this amplification.¹⁰ Their framework generates firm entry during booms, intensifying competition and compressing markups. This raises labor demand more than in a standard model. The resulting endogenous markup compression increases the labor share during expansions, making labor income more volatile relative to aggregate output. Since constrained households consume their labor income entirely, $\sigma_{c_{htm}}/\sigma_y$ rises above unity, directly amplifying the first term in the variance decomposition.

The empirical evidence on this mechanism is more limited than the evidence on financial access. Long cross-country panels that jointly measure firm entry, markups, and labor-share cyclicalities are not available for the full set of economies used in the estimation.¹¹ We therefore treat the firm-entry block as a disciplined amplification layer, supported by existing evidence on labor-share dynamics and entry during sudden stops, while allowing the macroeconomic data to inform the key production elasticities in the full model.

In what follows, we augment the baseline model with endogenous firm entry and variable markups, following [Jaimovich and Floetotto \(2008\)](#), layered directly on top of the baseline framework developed in Section 3. To maintain comparability with the baseline, all parameters are kept consistent with those reported in Table 2, with two exceptions: the capital share parameter α is adjusted to 0.11 to preserve a steady-state labor share of 65% in the presence of firm dynamics, and the standard deviation of the productivity shock σ_a is recalibrated to 0.0125 to maintain an output volatility of 3.3%. The extended model introduces three additional parameters specific to the [Jaimovich and Floetotto \(2008\)](#) framework: the fixed cost of production $\varphi = 0.1$, the within-sector elasticity of substitution $\tau = 0.85$, and the across-sector elasticity of substitution $\omega = 0.05$. Full details of the model equations and the complete set of parameters are provided in Appendix

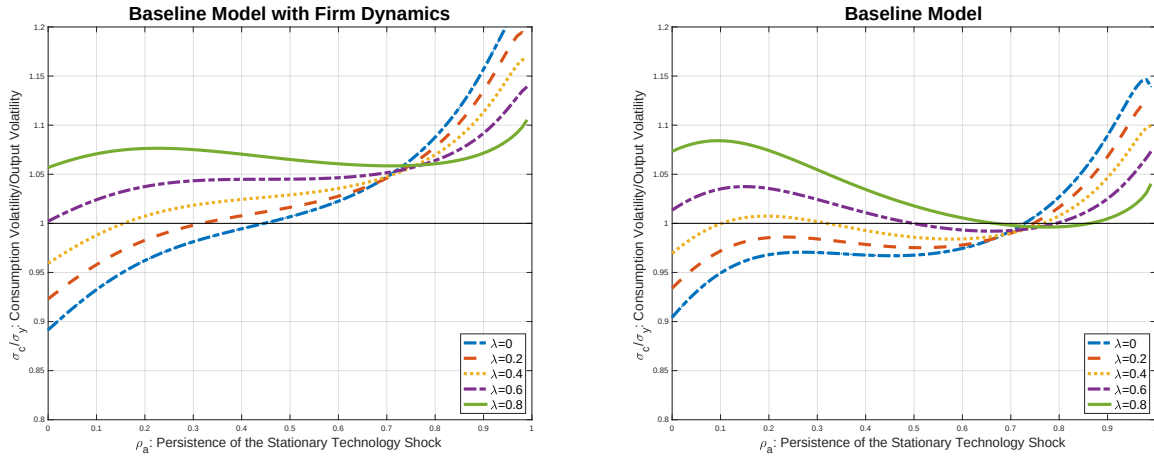
¹⁰[Ates and Saffie \(2021\)](#) confirm this pattern using plant-level data from Chile, showing that firm entry fell by 29 percent during the sudden stop. Related evidence and mechanisms for the United States are provided by [Rios-Rull and Santaella-Llopis \(2010\)](#) and [Choi and Rios-Rull \(2009\)](#).

¹¹For example, [Na \(2022\)](#) report labor-share cyclicalities for 26 countries, but only 3 of the 28 emerging and low-income economies in our estimation sample appear in Na's data. [Karabarbounis and Neiman \(2014\)](#) provide broader country coverage, but imposing a minimum of 20 continuous annual labor-share observations leaves only 16 usable countries, including 2 emerging economies and no low-income economies.

A.5 for reference. Using this augmented structure, we first examine whether household heterogeneity continues to amplify the relative volatility of consumption as documented in Figure 2, and then assess the extent to which procyclical firm dynamics strengthen the hand-to-mouth channel relative to the baseline.

Figure 4 presents relative consumption volatility as a function of shock persistence, ρ_a , for different shares of hand-to-mouth households, λ . The left panel shows the baseline model with firm dynamics, and the right panel shows the baseline model without firm dynamics. Two conclusions emerge from this comparison.

Figure 4: Ratio of consumption volatility and output volatility and the relationship with persistence of transitory/stationary technology shock in the model



Notes: This figure shows how changes in the persistence of the technology shock given by ρ_a (on the horizontal axis) affect the model-generated ratio of the volatility of consumption to the volatility of output given by $\text{SD}(c) / \text{SD}(y)$. The left panel shows the result for the augmented model while the right panel shows the same relationship for the baseline model, reproducing Figure 2. The figures are based on the parameterization summarized in Table 2, with $\alpha = 0.11$ and $\sigma_a = 0.0125$.

First, the model with firm dynamics successfully replicates the core finding of the baseline model: household heterogeneity raises the relative volatility of consumption and this relationship holds for a wide range of low and medium persistence values. The figure for the augmented model closely mirrors that of the baseline, confirming that the unconstrained amplification mechanism is robust to the introduction of endogenous firm entry and variable markups.

Second, the two models diverge at higher levels of persistence. In the baseline model, the relative consumption volatility curves for high- λ economies exhibit a non-monotonic relationship with ρ_a . In the model with firm dynamics, by contrast, this relationship

remains broadly monotonic across all values of λ . While the quantitative gain in relative consumption volatility (the primary focus of the paper) attributable to firm dynamics is modest, this has a favorable implication for identification in the estimation exercise of Section 4.1. The model with firm dynamics, therefore, serves as the preferred specification for estimation.¹²

To evaluate the quantitative gain from firm dynamics numerically, we return to the variance decomposition introduced earlier. Table 4 shows four configurations: the baseline model and the augmented model with firm dynamics, each evaluated at two levels of shock persistence, $\rho_a = 0.17$, using the calibrated value from Table 2, and $\rho_a = 0.7$, an intermediate value of persistence.

Comparing Panels A and B at the baseline persistence, the introduction of firm dynamics raises $\sigma_{c_{htm}}/\sigma_y$ from approximately 0.906 to 0.966 across all values of λ , a meaningful improvement, but one that falls short of pushing the hand-to-mouth relative volatility above unity. This explains why the quantitative gain in aggregate relative consumption volatility is modest at low persistence: while firm dynamics do strengthen the hand-to-mouth channel, the labor share mechanism is not operating at full force under a weakly persistent shock. When the shock dissipates quickly, the endogenous markup compression induced by firm entry is short-lived, limiting the extent to which labor income becomes more volatile than aggregate output. Consequently, aggregate relative consumption volatility is slightly lower in Panel B than in Panel A across most values of λ , as the modest improvement in the hand-to-mouth channel is partially offset by a small reduction in unconstrained household volatility $\sigma_{c_{unc}}/\sigma_y$.

The contrast between Panels C and D, evaluated at the higher persistence level, is considerably more striking and directly explains the divergence between the two models observed in Figure 4. In the baseline model (Panel C), $\sigma_{c_{htm}}/\sigma_y$ rises above unity, reaching a value close to 1.02. The unconstrained household volatility $\sigma_{c_{unc}}/\sigma_y$, however, falls to 1.01 at $\lambda = 0.2$ and 1.30 at $\lambda = 0.8$, a substantially compressed range relative to the low-persistence case. The net effect is that aggregate relative consumption volatility hovers just below unity across all values of λ in the baseline, explaining the downward turn of the baseline curves at high persistence in Figure 4. In the model with firm dynamics (Panel D), the picture changes markedly. Firm dynamics push $\sigma_{c_{htm}}/\sigma_y$ above unity more forcefully, reaching values close to 1.09, reflecting the full operation of the labor share channel under

¹²Specifically, advanced economies with a relative consumption volatility of unity can satisfy $\lambda = 0.2$ or $\lambda = 0.8$ for a wide and realistic range in the persistence of the productivity shock in the baseline model. This becomes a potential concern for the estimation.

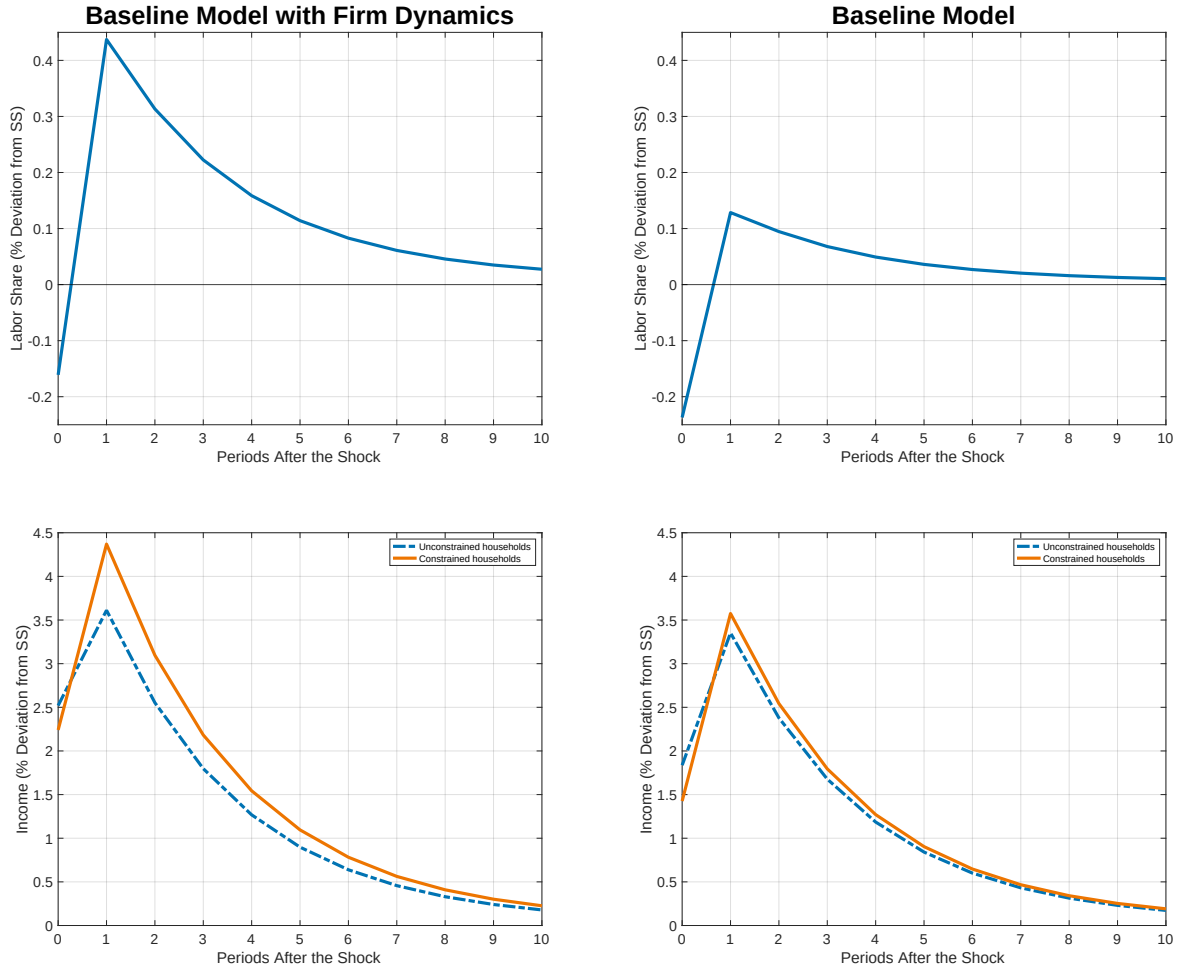
a more persistent productivity boom. Crucially, this amplified hand-to-mouth volatility is now sufficient to push aggregate relative consumption volatility above unity across all values of λ : the aggregate relative variance, σ_c^2/σ_y^2 , ranges from 1.09 to 1.12, corresponding to relative volatility values of roughly 1.05 to 1.06.

Figure 5 illustrates the mechanism behind these results through impulse response functions evaluated at a higher persistence level, $\rho_a = 0.70$, where this mechanism is stronger. The top panels show the labor share response and the bottom panels show the income responses for constrained and unconstrained households. The left panels correspond to the baseline model with firm dynamics, and the right panels correspond to the baseline model without firm dynamics.

Consider first the labor share IRFs. In both models, the labor share drops on impact, in accordance with the predetermined nature of capital as evident from equation 4. In the subsequent periods, from $t = 1$ onward, the labor share recovers and overshoots its steady-state value, consistent with the empirical evidence in [Rios-Rull and Santaaulalia-Llopis \(2010\)](#). The key difference between the two models lies in the magnitude and persistence of this overshoot. In the augmented model, procyclical firm entry during the productivity boom intensifies competition and compresses markups, further raising the demand for labor relative to capital. This amplifies the labor share overshoot from $t = 1$ onward, keeping it elevated for longer. Under a persistent shock, this amplification operates at full force: the productivity boom is sufficiently prolonged for the markup compression channel to build up, generating a labor share response that is visibly larger in the firm dynamics model than in the baseline throughout the recovery phase.

The income IRFs translate this directly into household-level incomes. In both models, constrained household income, which equals constrained household consumption, tracks the labor share closely: it undershoots unconstrained (and aggregate) income on impact but overtakes it from $t = 1$ onward. The firm dynamics model amplifies this crossover. Because the labor share overshoot is larger and more persistent, constrained household income deviates more from steady state over the medium run relative to unconstrained household income (and aggregate income). Since constrained households consume their entire income, this translates one-for-one into a larger consumption deviation. It is precisely this amplification that pushes $\sigma_{c_{htm}}/\sigma_y$ from 1.02 in the baseline to 1.09 in the augmented model at $\rho_a = 0.70$. Combined with the unconstrained amplification, this is sufficient to sustain aggregate relative consumption volatility above unity across all values of λ at empirically plausible levels of shock persistence. It is this property that makes the

Figure 5: Impulse responses to a 1% transitory productivity shock evaluated at a higher persistence level, $\rho_a = 0.70$.



Notes: Each panel shows impulse response functions at $\rho_a = 0.70$ and $\lambda = 0.65$. The left panels show results for the baseline model with firm dynamics, and the right panels show results for the baseline model without firm dynamics. Top panels show the labor share response (percentage deviation from steady state). Bottom panels show the income responses of constrained and unconstrained households (percentage deviation from steady state); constrained household income equals constrained household consumption. All remaining parameters follow Table 2, with $\alpha = 0.11$ and $\sigma_a = 0.0125$ in the left panels.

augmented model the preferred specification for the estimation exercise that follows.

4. Full model

After using the simple model to isolate the mechanisms driving high consumption volatility under household heterogeneity and transitory productivity shocks, we extend

the analysis to a full structural model following [Aguiar and Gopinath \(2007\)](#), [Garcia-Cicco et al. \(2010\)](#), and [Neumeyer and Perri \(2005\)](#).

The full model builds on the framework developed in Section 3.4. It incorporates the key shocks commonly studied in the international macroeconomics literature. These include transitory and permanent productivity shocks, a_t and g_t ; an interest rate shock entering through the debt-elastic interest rate, ξ_t ; and a working capital requirement, whereby firms must finance a fraction η of their wage bill prior to production. Because the model structure closely follows the framework described in Section 3.4, augmented with the additional shocks outlined above, we report the full set of equilibrium conditions in Appendix A.6.

Using this enriched framework, we estimate the model parameters with Bayesian methods for a panel of 38 countries. The estimation recovers the posterior distribution of parameters by combining prior beliefs with the likelihood of the observed data. The data consist of time series for four key macroeconomic aggregates: the growth rates of output, consumption, and investment, as well as the trade balance-to-output ratio. The primary objective is to examine the estimated fraction of hand-to-mouth households across countries. We compare this estimate across advanced, emerging, and low-income economies to assess whether the estimation yields economically meaningful results. We also estimate the production-side elasticities that govern the firm-entry block. Detailed country-level estimation results are provided in the Online Appendix.

4.1. Model solution and estimation

We solve the model by perturbation. Estimation is Bayesian, with parameters categorized into two groups: those calibrated from the existing literature and those estimated from the data. Relative to the framework in Section 3.4, the full model introduces seven additional parameters. Six govern the persistence and volatility of two new shocks, the permanent productivity shock and the interest rate shock, while the transitory productivity shock was already present in Section 3.4. The seventh is η , the fraction of the wage bill firms must finance as working capital. All parameters, including those carried over from Section 3.4, are either calibrated or estimated. We discuss each in turn.¹³

¹³To mitigate concerns about short-sample bias in business cycle analysis ([Garcia-Cicco et al., 2010](#)), we rely on long time series spanning at least 50 years. For most countries, the sample runs from 1961 to 2019. The full list of countries is provided in Table 8.

Calibrated parameters. Table 5 presents the nine calibrated parameters. Six are generic, applied uniformly across all countries, while three are country-specific. The generic parameters are identical to those used in Section 3.4. The country-specific parameters are the real interest rate (r^*), the steady-state debt-to-GDP ratio ($\bar{d}/\bar{y}$), and the GDP growth rate (g), each calibrated using historical averages.

Table 5: Calibrated parameters of the full model

Parameter		C/G	Value	Target
International interest rate	r^*	C	0.025	Average interest rate (1961-2019)
Growth rate in real GDP per capita	g	C	1.0151	Average growth rate (1961-2019)
Consumption intertemporal elasticity	γ	G	2	Mendoza (1991)
Labor supply elasticity	ν	G	1.4	Mendoza (1991)
Capital depreciation rate	δ	G	0.15	Standard
Production function capital share	α	G	0.11	Labor share of 65%
Elasticity of substitution capital/labor	ζ	G	0.6	Choi and Rios-Rull (2009)
Fixed cost in production	φ	G	0.1	Firm market share of 40%
External debt to output	$\bar{d}/\bar{y}$	C	-0.78	Average TBY (1961-2019)

Note: C/G indicates whether a variable is country-specific or generic. A generic variable takes the same value for all countries. Country-specific variables differ by country. The values reported for country-specific variables pertain to Mexico.

To fix ideas, consider Mexico as an illustrative example. The real interest rate is set at $r^* = 2.5\%$, computed from WDI data. The steady-state debt-to-GDP ratio is matched to the country's average trade balance-to-output ratio using $\bar{d}/\bar{y} = tby/(1 - g/(1 + r^*))$, yielding $\bar{d}/\bar{y} = -0.78$, consistent with an average trade balance of -0.75% of GDP. Finally, the average per capita GDP growth rate of 1.51% implies a calibrated value of $g = 1.0151$.

Estimated parameters. Table 6 presents the priors and posteriors for the 12 structural and 4 non-structural parameters estimated via Bayesian methods for each country. The structural parameters comprise the persistence and volatility of the three shocks ($\rho_a, \rho_g, \rho_\xi, \sigma_a, \sigma_g, \sigma_\xi$), the investment adjustment cost parameter (ϕ), the debt elasticity of the interest rate (ψ), the working capital constraint parameter (η), the across-sector elasticity of substitution (ω), the within-sector elasticity of substitution (τ), and the fraction of hand-to-mouth households (λ). The four non-structural parameters, σ_{gY}^{me} , σ_{gC}^{me} , σ_{gInv}^{me} , and σ_{tby}^{me} ,

Table 6: Estimated parameters of the full model

Parameter	Prior			Posterior		
	Min.	Max.	Distribution	Advanced	Emerging	Low-income
σ_a	0.0000	0.2000	Uniform	0.0113	0.0229	0.0190
ρ_a	-0.9900	0.9900	Uniform	0.8587	0.8211	0.7340
σ_g	0.0000	0.2000	Uniform	0.0041	0.0215	0.0188
ρ_g	-0.9900	0.9900	Uniform	0.9778	0.9562	0.9844
σ_ξ	0.0000	0.2000	Uniform	0.0159	0.0459	0.0680
ρ_ξ	-0.9900	0.9900	Uniform	0.6912	0.8073	0.8378
ϕ	0.0000	5.0000	Uniform	0.8830	1.7762	1.8988
ψ	0.0000	2.0000	Uniform	0.4113	0.3730	0.3600
η	0.0000	1.0000	Uniform	0.4107	0.3577	0.2714
ω	0.0000	0.5000	Uniform	0.2912	0.3241	0.2488
τ	0.5000	1.0000	Uniform	0.9496	0.8564	0.9440
$\sigma_{g_Y}^{me}$	0.0000	$0.25 \cdot std(g_Y)$	Uniform	0.0033	0.0080	0.0077
$\sigma_{g_C}^{me}$	0.0000	$0.25 \cdot std(g_C)$	Uniform	0.0048	0.0108	0.0123
$\sigma_{g_{Inv}}^{me}$	0.0000	$0.25 \cdot std(g_{Inv})$	Uniform	0.0163	0.0332	0.0443
σ_{tby}^{me}	0.0000	$0.25 \cdot std(tby)$	Uniform	0.0019	0.0077	0.0079
λ	0.0000	0.9900	Uniform	0.1600	0.5055	0.5865

Note: Mean values for all parameters are taken for each country and the mean across the group of advanced, emerging and low-income countries is reported. Country-wise results are reported in Table 10 of the Online Appendix.

capture measurement error in the observed macroeconomic series.

All priors are uniform. Shock persistence parameters are bounded between -0.99 and 0.99 , and shock volatilities between 0 and 0.2 . The investment adjustment cost (ϕ) is restricted to $[0,5]$, the debt elasticity (ψ) to $[0,2]$, and the working capital fraction (η) to $[0,1]$. The bounds on the two elasticity parameters, ω and τ , are informed by the existing literature. For the across-sector elasticity, ω , we follow [Backus et al. \(1994\)](#), who calibrate an Armington elasticity of 1.5 , and allow the parameter to range over values consistent with an elasticity between 1 and 2 , implying bounds of 0 and 0.5 on ω . For the within-sector elasticity, τ , we draw on [Broda and Weinstein \(2006\)](#), whose estimates range from 3 to 12.6 across goods categories; we set the bounds to $[0.5,1]$ to span this range and allow for some additional flexibility. The measurement error parameters are constrained so that measurement error accounts for no more than 6.25% of the variance of each observable.

Finally, the fraction of hand-to-mouth households λ is bounded between 0 and 0.99.

4.2. Model results

Estimated parameters. Table 6 reports the posterior means of all estimated parameters, averaged across countries within each income group. We discuss the results in turn, with particular emphasis on the fraction of hand-to-mouth households λ .

The shock parameters reveal intuitive patterns across income groups. The volatility of the transitory productivity shock, σ_a , rises from advanced (0.011) to emerging (0.023) and low-income economies (0.019), consistent with the well-documented finding that output fluctuations are larger in emerging and less developed economies. The permanent productivity shock is similarly more volatile in emerging and low-income economies. The interest rate shock volatility, σ_ξ , increases sharply across groups, from 0.016 in advanced economies to 0.068 in low-income economies, reflecting the greater exposure of developing economies to external financial conditions. Shock persistence is broadly similar across groups.

The investment adjustment cost, ϕ , is substantially lower in advanced economies (0.883) than in emerging (1.776) and low-income (1.899) economies, suggesting that capital reallocation is less costly in advanced economies. The debt elasticity, ψ , is broadly similar across groups, hovering around 0.37, which may reflect the fact that the debt-elastic interest rate specification captures a similar equilibrating mechanism regardless of development level. The working capital parameter η is somewhat higher in advanced economies (0.411) than in emerging (0.358) and low-income (0.271) economies, a pattern that warrants caution in interpretation, as working capital constraints are typically associated with less financially developed economies. One possible explanation is that advanced economies have larger and more capital-intensive firms that require greater working capital financing even in the absence of financial frictions.

Turning to the elasticity parameters, the posterior means of the across-sector elasticity, ω , and the within-sector elasticity, τ , are broadly consistent across income groups. The estimated values of τ , ranging from 0.856 to 0.950, are close to the value of 0.85 used in [Jaimovich and Floetotto \(2008\)](#). The estimated values of ω , ranging from 0.249 to 0.324, are higher than the value of 0.05 used in that calibration but remain inside the broader range implied by the Armington-elasticity priors discussed above. We therefore interpret these estimates not as a replication of the [Jaimovich and Floetotto \(2008\)](#) calibration, but as evidence that the macroeconomic data are informative about the production-side margin.

The most salient finding concerns the fraction of hand-to-mouth households λ . The posterior mean rises sharply and monotonically across income groups: 0.160 for advanced, 0.506 for emerging, and 0.587 for low-income economies. This pattern is both statistically and economically meaningful. It implies that roughly one in six households in advanced economies are hand-to-mouth, compared to approximately one in two in emerging economies and nearly three in five in low-income economies. These estimates are broadly consistent with the microeconomic evidence on the prevalence of liquidity-constrained households (Kaplan and Violante, 2014; Honohan, 2008). Importantly, the estimated gradient in λ across income groups provides direct support for the central mechanism of the model: household heterogeneity is not merely a theoretical device but a quantitatively important feature of business cycles, particularly in emerging and low-income economies.

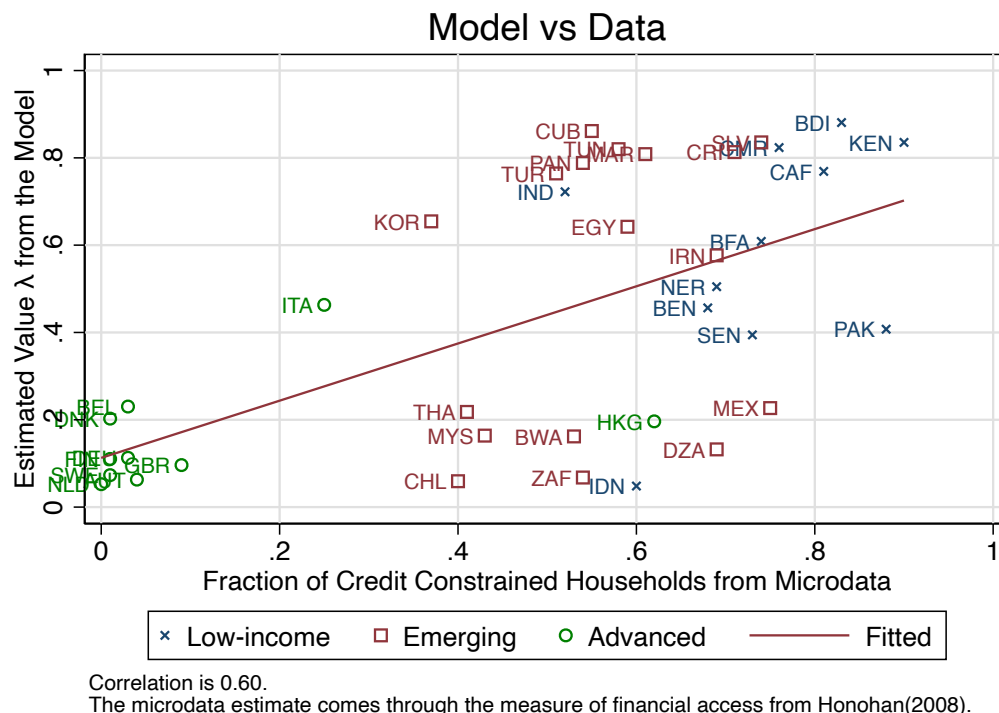
Fraction of credit-constrained households: macrodata vs microdata. Among the estimated parameters, the fraction of hand-to-mouth households λ speaks directly to the degree of financial inclusion across economies. The group-level posterior means follow an intuitive gradient: 16% in advanced, 51% in emerging, and 59% in low-income economies. While financial markets have improved substantially in many developing economies in recent decades, these estimates reflect average conditions over the full 1961–2019 sample.

To validate these estimates, we compare them against the microdata evidence of Honohan (2008), who measure financial access using household surveys and bank account records. Figure 6 plots the model-implied λ against the fraction of hand-to-mouth households derived from their data for all 38 countries. The correlation of 0.60 is encouraging: despite relying on only three shocks and four macroeconomic time series, with no direct information on household balance sheets, the model recovers estimates of λ that align closely with independent microdata evidence.

Advanced economies cluster near the lower end of both axes and lie close to the fitted line. Emerging economies display the greatest dispersion, reflecting the wide variation in financial development within this group. Low-income economies align reasonably well with the fitted line, with the exception of Indonesia.

We assess the robustness of this finding in two ways, with results reported in Figures 7 and 8 in Appendix A.3. First, we calibrate ω and τ to the values of Jaimovich and Floetotto (2008) and re-estimate the remaining 14 parameters. Second, we shut down firm dynamics and move the production structure closer to Cobb-Douglas by increasing ζ from 0.6 to 0.8, again re-estimating 14 parameters. In both cases, the posterior means of λ remain

Figure 6: Estimated fraction of credit-constrained households and microdata counterpart



close to the benchmark estimates, preserving the monotonic gradient across income groups. Overall, the alignment between macro- and micro-based measures, together with the robustness of the gradient in λ , lends credibility to the model and supports the interpretation of λ as a meaningful indicator of household financial constraints.

5. Policy implications

The results point to a direct implication for stabilization policy. Aggregate shocks are transmitted through a distributional margin: a large fraction of households cannot smooth disposable income, while unconstrained households absorb capital-income and external-financing risk. Policies that stabilize aggregate absorption need not stabilize private consumption unless they insure the households whose consumption moves one-for-one with current income.

To make this point transparent, consider two zero-steady-state fiscal rules added to

the baseline model in Section 3. The fiscal incidence assumptions are deliberately simple. Public absorption is non-valued and is financed contemporaneously, so an increase in this type of expenditure reduces households' disposable resources rather than providing direct utility. The broad absorption rule is therefore an aggregate resource stabilization instrument. By contrast, the targeted-transfer rule changes the composition of public spending toward households without financial access. The exercise is not a welfare comparison. It isolates whether stabilization works through the disposable-income margin that drives the mechanism.

For the first policy, broad public absorption follows

$$\Delta G_t^A = -\tau_g \bar{y} (\log y_t - \log \bar{y}), \quad (6)$$

where $\tau_g > 0$ raises public absorption in recessions and lowers it in booms. This is a standard aggregate stabilization instrument, but it is not targeted to households without financial access.

For the second policy, each hand-to-mouth household receives a targeted transfer

$$T_t^{htm} = -\tau_T (1 - \lambda) \bar{y} (\log y_t - \log \bar{y}), \quad (7)$$

which is positive in recessions. Because the aggregate cost is λT_t^{htm} , the policy finances the transfer by reducing non-transfer public absorption one-for-one:

$$\Delta G_t^T = -\lambda T_t^{htm}, \quad \Delta G_t^T + \lambda T_t^{htm} = 0. \quad (8)$$

Thus, the targeted rule is a budget-neutral composition switch within public expenditure. The resources used for transfers are exactly the resources released by lower non-valued public absorption, without relying on public debt or additional taxes. Under these incidence assumptions, hand-to-mouth disposable income equals labor income net of the broad absorption claim plus targeted transfers:

$$c_t^{htm} = w_t h_t - \Delta G_t^A + T_t^{htm}, \quad (9)$$

and the aggregate resource constraint is

$$c_t = y_t - i_t - \Delta G_t^A - \Delta G_t^T - \frac{\phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t + \frac{d_{t+1}}{1 + r_t} - d_t. \quad (10)$$

Table 7 reports illustrative moments under the same structural calibration as the simple model in Table 2, with only the productivity shock active. The countercyclical-absorption rule uses $\tau_g = 0.05$, and the targeted-transfer rule uses $\tau_T = 0.6$. These exercises are not optimal-policy calculations. They instead isolate the margin the model identifies as relevant for consumption volatility.

Table 7: Illustrative fiscal rules in the simple model

Policy experiment	σ_y	σ_c/σ_y	$\sigma(y_{disp}^{htm})$	$\sigma(y_{disp}^{unc})$
<i>Low hand-to-mouth share, $\lambda = 0.10$</i>				
Baseline	3.33	0.97	3.02	3.35
Countercyclical ΔG_t^A	3.34	1.05	3.29	3.53
Budget-neutral targeted T_t^{htm}	3.33	0.91	0.37	3.35
<i>High hand-to-mouth share, $\lambda = 0.65$</i>				
Baseline	3.33	1.05	3.02	3.57
Countercyclical ΔG_t^A	3.33	1.11	3.28	3.68
Budget-neutral targeted T_t^{htm}	3.33	0.89	1.95	3.57

Notes: Volatilities are standard deviations of log deviations from steady state, in percent, except for σ_c/σ_y . The rules set $\tau_g = 0.05$ and $\tau_T = 0.6$. Targeted transfers are budget neutral, with $\Delta G_t^T = -\lambda T_t^{htm}$. Disposable income of hand-to-mouth households coincides with their consumption.

Two lessons follow from these results. First, broad countercyclical public absorption is a weak instrument for the mechanism emphasized in this paper: it raises both relative private consumption volatility and hand-to-mouth disposable-income volatility. The reason is that higher public absorption in recessions uses resources precisely when constrained households are most exposed to low current labor income. Without a targeted transfer, that instrument does not relax their disposable-income constraint.

Second, the budget-neutral targeted transfer directly insures the relevant households. With $\lambda = 0.65$, it reduces hand-to-mouth disposable-income volatility from 3.02 percent to 1.95 percent and lowers relative aggregate consumption volatility from 1.05 to 0.89. With $\lambda = 0.10$, it reduces hand-to-mouth disposable-income volatility from 3.02 percent to 0.37 percent and lowers relative aggregate consumption volatility from 0.97 to 0.91. The gain comes from reallocating non-valued public absorption toward households whose consumption is tied to current income.

A complementary interpretation is that stabilization policy may operate by reducing inefficient labor-demand fluctuations. In New Keynesian environments with nominal

rigidities, cyclical markups or demand externalities can appear as labor wedges, in the sense of business-cycle accounting. In the present model, such wedges matter primarily because they alter the labor income of hand-to-mouth households. Hence, policies that stabilize inefficient movements in labor demand can reduce consumption volatility when they stabilize constrained households' disposable income. This interpretation reinforces, rather than replaces, the fiscal-incidence message: the relevant policy object is the cyclical exposure of constrained households' current income.

Overall, the model cautions against interpreting excess consumption volatility as only an aggregate-shock problem. Stabilization tools that ignore household financial access can leave the mechanism intact, whereas budget-neutral policies that redirect existing fiscal resources toward constrained households are better aligned with the source of volatility identified in the paper.

6. Conclusion

This paper investigates the role of household heterogeneity in financial access as a driver of excess consumption volatility in emerging and low-income economies. Standard macroeconomic models, when subject to only transitory productivity shocks, predict consumption smoothing through financial markets, yet empirical evidence consistently shows that aggregate consumption is more volatile than income in these economies. Our framework shows that hand-to-mouth households can generate excess aggregate consumption volatility not only through mechanical aggregation, but primarily through the unconstrained-amplification channel: as capital income is concentrated among fewer unconstrained households, their consumption response unwinds more slowly after a transitory shock, raising its unconditional volatility relative to output.

Procyclical firm dynamics provide a second, complementary amplification layer. By compressing markups and raising the labor share during expansions, firm entry makes labor income more volatile relative to aggregate output and strengthens the hand-to-mouth channel. The mechanism is therefore distributional in two senses: shocks affect labor and capital income differently, and financial access determines which households can smooth those income fluctuations.

Through Bayesian estimation on a dataset of 38 countries spanning advanced, emerging, and low-income economies, we quantify the fraction of hand-to-mouth households and estimate the production-side elasticities that govern the firm-entry block. The estimated

fraction of hand-to-mouth households rises monotonically with lower levels of development and aligns closely with microdata on financial access, lending credibility to the macroeconomic interpretation of this parameter. The policy implication is that aggregate stabilization alone need not stabilize private consumption: policies are most aligned with the mechanism when they insure the current disposable income of households without access to financial markets.

Data Availability

The data and code supporting the findings of this study are available in the Harvard Data-verse replication package (Singh and Tavares, 2026), at <https://doi.org/10.7910/DVN/DA0C4D>.

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A. Appendix

Appendix A collects the supporting material used in the main text. Appendix A.1 documents the construction of the empirical moments and the financial-access measure. Appendix A.2 reports the comparison with the global solution method, and Appendix A.3 reports additional figures for the estimated hand-to-mouth shares. The Online Appendix then contains the complete model specifications and the country-level estimation results.

A.1. Additional details on the empirical motivation

This section provides additional details on the sample selection, data, and statistics shown in Section 2 of the main text.

The business cycle statistics use annual national accounts data from the 2023 edition of the *World Development Indicators*, compiled by the *World Bank*. We retain countries with at least 50 years of continuous data for output, total consumption, investment, and the trade balance, and restrict attention to economies with populations above one million. All variables are expressed in local currency at constant prices. The cyclical component is extracted from log variables using an HP-filter, except for the trade-balance-to-output ratio, which is expressed as a deviation from its trend. Group-level moments are weighted by each country's share of world output. The final sample contains 38 countries: 10 developed, 17 emerging, and 11 low-income economies. In Table 1, emerging and low-income economies are pooled in the column labeled “Emerging.”

The financial-access measure comes from Honohan (2008), who estimates the fraction of the adult population using formal financial intermediaries from household surveys, bank account data, and microfinance information. We use this measure to proxy for the inverse of the fraction of hand-to-mouth households in the empirical motivation and to discipline the calibration of λ in the simple model.

Data sources and variable definition. The national accounts data used to compute business cycle statistics are sourced from the *World Development Indicators* dataset. Specifically, we select the following variables:

Output: Country yearly gross domestic product from the national accounts measured in constant local currency (code NY.GDP.MKTP.KN).

Total consumption: Aggregate total final consumption expenditure from the national accounts measured in constant local currency (code NE.CON.TOTL.KN).

Private consumption: Aggregate household final consumption expenditure from the national accounts measured in constant local currency (code NE . CON . PRVT . KN).

Public consumption: Aggregate central government final consumption expenditure from the national accounts measured in constant local currency (code NE . CON . GOVT . KN).

Investment: Aggregate gross fixed capital formation from the national accounts measured in constant local currency (code NE . GDI . FTOT . KN).

Trade balance to output: This variable is defined as exports of goods and services (code NE . EXP . GNFS . KN) minus imports of goods and services (code NE . IMP . GNFS . KN), divided by gross domestic product (code NY . GDP . MKTP . KN), with all series measured in constant local currency.

Interest rate: The real interest rate is the lending interest rate adjusted for inflation as measured by the GDP deflator (code FR . INR . RINR).

Country classification into developed, emerging, and low-income economies. The data for macroeconomic variables use the *World Development Indicators* dataset and the country classification follows the definitions provided by the World Bank in 2023. In particular, countries are included in the developed-economy group if their 2022 GNI per capita in purchasing power parity (Atlas method) is larger than \$13,205. Countries are included in

Table 8: Country classification according to income level

Country	Total
Developed economies:	10
Austria (AUT); Belgium (BEL); Denmark (DNK); Finland (FIN); Germany (DEU); Hong Kong SAR, China (HKG); Italy (ITA); Netherlands (NLD); Sweden (SWE); United Kingdom (GBR)	
Emerging economies	17
Algeria (DZA); Botswana (BWA); Chile (CHL); Costa Rica (CRI); Cuba (CUB); Egypt, Arab Rep. (EGY); El Salvador (SLV); Iran, Islamic Rep. (IRN); Korea, Rep (KOR); Malaysia (MYS); Mexico (MEX); Morocco (MAR); Panama (PAN); South Africa (ZAF); Thailand (THA); Tunisia (TUN); Turkey (TUR)	
Low-income economies	11
Benin (BEN); Burkina Faso (BFA); Burundi (BDI); Cameroon (CMR); Central African Republic (CAF); India (IND); Indonesia (IDN); Kenya (KEN); Niger (NER); Pakistan (PAK); Senegal (SEN)	

Source: *World Development Indicators* and *World Bank*.

the emerging-economy group if GNI per capita is between \$1,086 and \$13,205, and in the low-income-economy group if GNI per capita is below \$1,086. Table 8 specifies the list of countries in each group.

A.2. Robustness results: global methods

This section reports the robustness exercise that solves the simple model with a global method instead of perturbation. Table 9 compares the resulting business-cycle moments with the data and with the perturbation solution used in the baseline analysis. The model specification used for the global-method exercise is described in detail in the Online Appendix, Appendix A.4.4.

Table 9: Business cycle moments: data vs model

Moments		Data	Baseline model	
			Perturbation $\lambda = 0.65$	Global method $\lambda = 0.65$
Targeted moments				
Standard deviation of output	$\sigma(y)$	3.3	3.3	3.7
Autocorrelation of output	$\rho(y)$	0.48	0.53	0.51
Relative standard deviation:				
Investment	$\sigma(i)/\sigma(y)$	3.20	3.24	3.15
Trade balance	$\sigma(tby)/\sigma(y)$	0.72	0.72	0.74
Non-targeted moments				
Relative standard deviation:				
Consumption	$\sigma(c)/\sigma(y)$	1.04	1.05	1.06
Correlation with output:				
Consumption	$\rho(y, c)$	0.74	0.85	0.84
Investment	$\rho(y, i)$	0.67	0.50	0.28
Trade balance	$\rho(y, tby)$	-0.23	-0.09	0.13

Notes: The column ‘Perturbation’ shows the moments generated by the baseline model, solved using the perturbation method and parameterized as in Table 2. The column ‘Global method’ shows the moments generated by the baseline model solved using the global solution method. In both cases, the fraction of financially constrained households is the same as in the baseline version of the model, $\lambda = 0.65$.

A.3. Additional Figures

This section reports two robustness figures for the estimated hand-to-mouth share. Figure 7 compares the model-implied λ with the microdata counterpart when ω and τ are calibrated, while Figure 8 repeats the comparison after shutting down the firm-dynamics channel.

Figure 7: Estimated fraction of credit-constrained households and microdata counterpart, with ω and τ calibrated

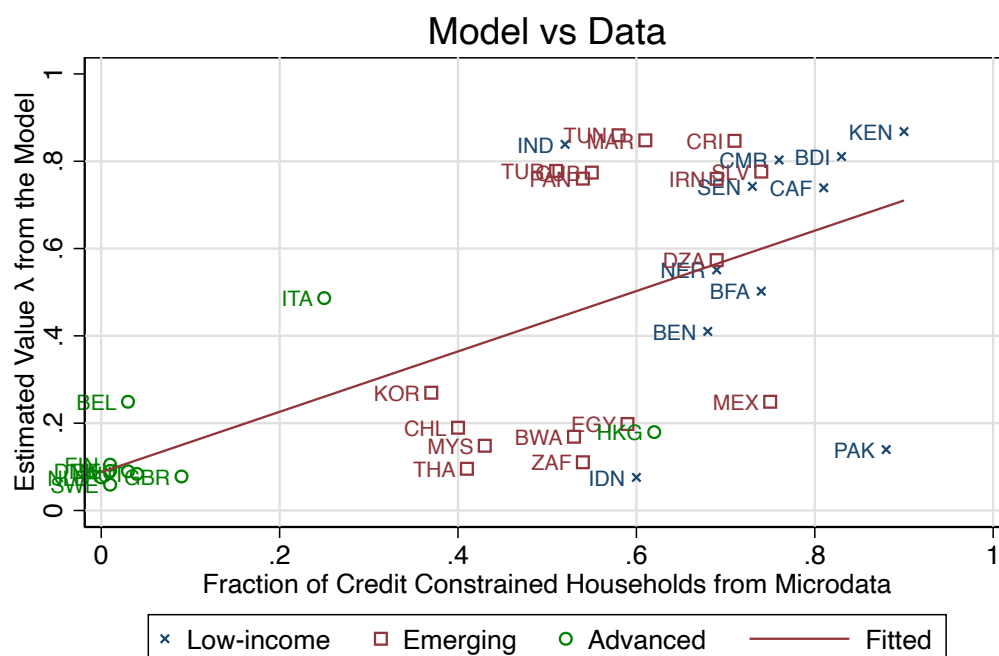
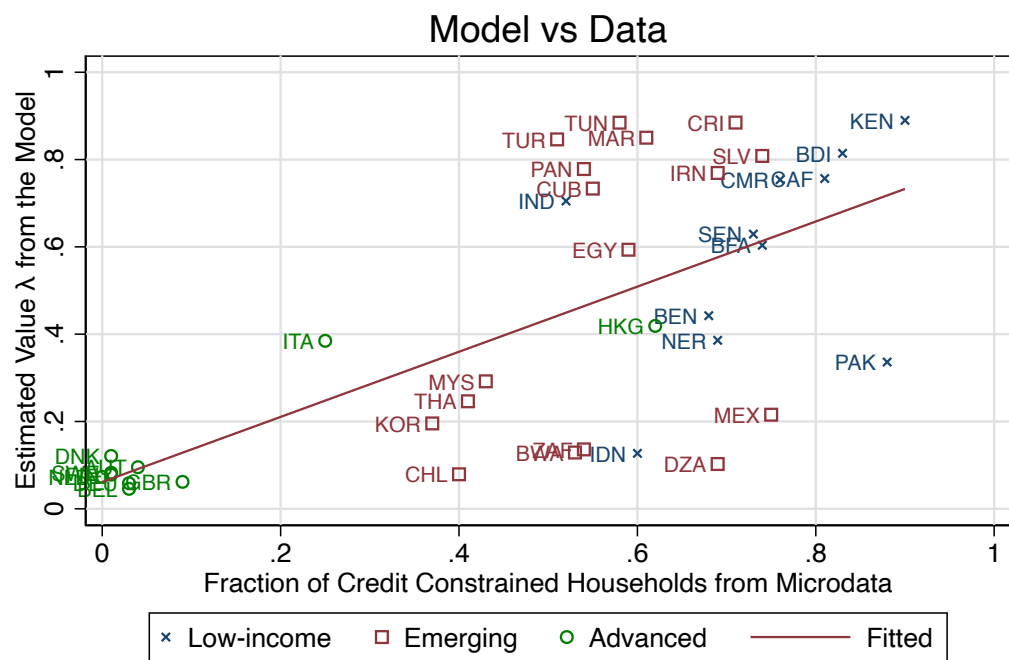


Figure 8: Estimated fraction of credit-constrained households and microdata counterpart, without the firm-dynamics channel



Correlation is 0.67.
The microdata estimate comes through the measure of financial access from Honohan(2008).

Online Appendix

The Online Appendix collects the technical material that supports the quantitative analysis in the main text. It reports the full equilibrium conditions and steady states for the simple model, the version with firm dynamics, and the full estimated model, followed by robustness exercises and additional estimation results.

A.4. Complete specification of the simple model

This section states the complete set of equations behind the simple model used in Section 3. It first lists the baseline equilibrium conditions, then describes the fiscal-policy extension, the steady state, and the formulation used for the global-solution exercise.

A.4.1. Baseline equilibrium conditions

The baseline economy has two household types and a representative firm. The equations below state the household optimality conditions, the firm's static conditions, market clearing, and the exogenous shock process used in the simple model.

Hand-to-mouth households.

$$c_t^{htm} = w_t h_t^{htm} \quad (11)$$

$$w_t = (h_t^{htm})^{\nu-1} \quad (12)$$

Unconstrained households.

$$\left(c_t^{unc} - \frac{(h_t^{unc})^\nu}{\nu} \right)^{-\gamma} = \zeta_t^{unc} \quad (13)$$

$$w_t = (h_t^{unc})^{\nu-1} \quad (14)$$

$$\zeta_t^{unc} = \beta(1 + r_t)E_t[\zeta_{t+1}^{unc}] \quad (15)$$

$$\zeta_t^{unc} \left(1 + \phi \left(\frac{k_{t+1}}{k_t} - 1 \right) \right) = \beta E_t \left[\zeta_{t+1}^{unc} \left\{ 1 - \delta + r_{kt+1} + \frac{\phi}{2} \left(\left(\frac{k f u_{t+1}}{k f u_t} \right)^2 - 1 \right) \right\} \right] \quad (16)$$

$$c_t^{unc} = w_t h_t^{unc} + r_{kt} k_t^{unc} - i_t^{unc} - \frac{\phi}{2} \left(\frac{k_{t+1}^{unc}}{k_t^{unc}} - 1 \right)^2 k_t^{unc} + \frac{d_{t+1}^{unc}}{1 + r_t} - d_t^{unc} \quad (17)$$

Production.

$$\tilde{y}_t = a_t \left(\alpha \tilde{k}_t^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \tilde{l}_t^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (18)$$

$$r_{kt} = \alpha a_t \left(\frac{y_t}{a_t \tilde{k}_t} \right)^{\frac{1}{\zeta}} \quad (19)$$

$$w_t = (1 - \alpha) a_t \left(\frac{y_t}{a_t \tilde{l}_t} \right)^{\frac{1}{\zeta}} \quad (20)$$

Market clearing.

$$k_t = \tilde{k}_t \quad (21)$$

$$h_t = \tilde{l}_t \quad (22)$$

Aggregation.

$$c_t = \lambda c_t^{htm} + (1 - \lambda) c_t^{unc} \quad (23)$$

$$k_t = (1 - \lambda) k_t^{unc}, \quad i_t = (1 - \lambda) i_t^{unc}, \quad d_t = (1 - \lambda) d_t^{unc}. \quad (24)$$

Aggregate resource constraint.

$$c_t + k_{t+1} - (1 - \delta) k_t + \frac{\phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t + d_t = \frac{d_{t+1}}{1 + r_t} + y_t \quad (25)$$

Closing the model.

$$\log a_{t+1} = \rho_a \log a_t + \sigma_a \epsilon_{at+1} \quad (26)$$

$$r_t = r^* + \psi(e^{\frac{d_{t+1}-\bar{d}}{y}} - 1) \quad (27)$$

Other variable definitions.

$$kfu_t = k_{t+1} \quad (28)$$

$$I_t = k_{t+1} - (1 - \delta)k_t \quad (29)$$

$$sh_t^h = \frac{w_t h_t}{\tilde{y}_t} \quad (30)$$

$$tby_t = \frac{d_t - \frac{d_{t+1}}{1+r_t}}{\tilde{y}_t} \quad (31)$$

A.4.2. Fiscal policy extension

The fiscal exercises in Section 5 add government absorption and targeted transfers to the same baseline economy. Both policies are neutral in steady state, so the steady-state expressions below are unchanged.

Hand-to-mouth households.

$$c_t^{htm} = w_t h_t^{htm} - \Delta G_t^A + T_t^{htm} \quad (32)$$

$$w_t = (h_t^{htm})^{\nu-1} \quad (33)$$

Unconstrained households.

$$\left(c_t^{unc} - \frac{(h_t^{unc})^\nu}{\nu} \right)^{-\gamma} = \zeta_t^{unc} \quad (34)$$

$$w_t = (h_t^{unc})^{\nu-1} \quad (35)$$

$$\zeta_t^{unc} = \beta(1 + r_t)E_t[\zeta_{t+1}^{unc}] \quad (36)$$

$$\zeta_t^{unc} \left(1 + \phi \left(\frac{k_{t+1}}{k_t} - 1 \right) \right) = \beta E_t \left[\zeta_{t+1}^{unc} \left\{ 1 - \delta + r_{kt+1} + \frac{\phi}{2} \left(\left(\frac{k f u_{t+1}}{k f u_t} \right)^2 - 1 \right) \right\} \right] \quad (37)$$

$$c_t^{unc} = w_t h_t^{unc} + r_{kt} k_t^{unc} - i_t^{unc} - \frac{\phi}{2} \left(\frac{k_{t+1}^{unc}}{k_t^{unc}} - 1 \right)^2 k_t^{unc} + \frac{d_{t+1}^{unc}}{1 + r_t} - d_t^{unc} - \Delta G_t^A \quad (38)$$

Production.

$$\tilde{y}_t = a_t \left(\alpha \tilde{k}_t^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \tilde{l}_t^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (39)$$

$$r_{kt} = \alpha a_t \left(\frac{y_t}{a_t \tilde{k}_t} \right)^{\frac{1}{\zeta}} \quad (40)$$

$$w_t = (1 - \alpha) a_t \left(\frac{y_t}{a_t \tilde{l}_t} \right)^{\frac{1}{\zeta}} \quad (41)$$

Market clearing.

$$k_t = \tilde{k}_t \quad (42)$$

$$h_t = \tilde{l}_t \quad (43)$$

Aggregation.

$$c_t = \lambda c_t^{htm} + (1 - \lambda) c_t^{unc} \quad (44)$$

Aggregate resource constraint.

$$c_t + k_{t+1} - (1 - \delta) k_t + \frac{\phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t + d_t + \Delta G_t^A + \Delta G_t^T = \frac{d_{t+1}}{1 + r_t} + y_t \quad (45)$$

Incomes.

$$y_t^{htm} = c_t^{htm} \quad (46)$$

$$y_t^{unc} = w_t h_t + \frac{r_{kt} k_t}{1 - \lambda} - \Delta G_t^A \quad (47)$$

Absorption stabilizing fiscal policy (neutral in steady state).

$$\text{Countercyclical public absorption: } \Delta G_t^A = -\tau_g \bar{y} (\log(\tilde{y}_t) - \log(\bar{y})) \quad (48)$$

$$\text{Targeted transfer rule: } T_t^{htm} = -\tau_T (1 - \lambda) \bar{y} (\log(\tilde{y}_t) - \log(\bar{y})) \quad (49)$$

$$\Delta G_t^T = -\lambda T_t^{htm}, \quad \Delta G_t^T + \lambda T_t^{htm} = 0. \quad (50)$$

Closing the model.

$$\log a_{t+1} = \rho_a \log a_t + \sigma_a \epsilon_{at+1} \quad (51)$$

$$r_t = r^* + \psi \left(e^{\frac{d_{t+1} - \bar{d}}{\bar{y}}} - 1 \right) \quad (52)$$

Other variable definitions.

$$kfu_t = k_{t+1} \quad (53)$$

$$I_t = k_{t+1} - (1 - \delta)k_t \quad (54)$$

$$sh_t^h = \frac{w_t h_t}{\tilde{y}_t} \quad (55)$$

$$tby_t = \frac{d_t - \frac{d_{t+1}}{1+r_t}}{\tilde{y}_t} \quad (56)$$

A.4.3. Steady state

In the steady state with no shock ($\epsilon_{at} = 0, \forall t$)

$$a_{ss} = 1$$

$$r_{ss} = r^*$$

$$\beta(1 + r_{ss}) = 1$$

$$r_{kss} = \delta + r_{ss}$$

$$(k/h)_{ss} = \left(\frac{1 - \alpha}{\left(\frac{r_{kss}}{\alpha}\right)^{\zeta-1} - \alpha} \right)^{\frac{\zeta}{\zeta-1}}$$

$$h_{ss} = \left((1 - \alpha) \left(\alpha \left((k/h)_{ss} \right)^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \right)^{\frac{1}{\zeta-1}} \right)^{\frac{1}{\nu-1}}$$

$$k_{ss} = h_{ss} \cdot (k/h)_{ss}$$

$$Y_{ss} = \left(\alpha k_{ss}^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) h_{ss}^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}}$$

$$C_{ss}^{htm} = h_{ss}^{\nu}$$

$$Y_{ss}^{htm} = C_{ss}^{htm}$$

$$Y_{ss}^{unc} = \frac{Y_{ss} - \lambda Y_{ss}^{htm}}{1 - \lambda}$$

$$C_{ss}^{unc} = Y_{ss}^{unc} - \frac{\delta k_{ss}}{1 - \lambda} - \frac{r_{ss} \bar{d}}{(1 + r_{ss})(1 - \lambda)}$$

$$C_{ss} = \lambda C_{ss}^{htm} + (1 - \lambda) C_{ss}^{unc}$$

$$k f u_{kss} = k_{ss}$$

$$I_{ss} = \delta k_{ss}$$

$$sh_{ss}^h = \frac{w_{ss}h_{ss}}{Y_{ss}}$$

$$tby_{ss} = \frac{r_{ss}\bar{d}}{(1+r_{ss})Y_{ss}}$$

$$\zeta_{ss}^{unc} = \left(C_{ss}^{unc} - \frac{(h_{ss})^\nu}{\nu} \right)^{-\gamma}$$

A.4.4. Equilibrium conditions for global methods

We first express all variables in the model as functions of state and choice variables, and then perform value function iteration over the state-variable grids.

To begin, we use the labor supply and demand conditions to obtain labor as a function of capital, a state variable.

$$h_t^{\gamma-1} = (1-\alpha)a_t[\alpha(k_t/h_t)^{\frac{\zeta-1}{\zeta}} + (1-\alpha)]^{\frac{1}{\zeta-1}} \quad (57)$$

Because h_t cannot be solved analytically, we use a vectorized approach in MATLAB with `fminbnd`, constrained to positive values of h_t .

The next step is to write the remaining equations in recursive form and define the grids for the state variables.

Starting with the output:

$$y_t = a_t[\alpha k_t^{\frac{\zeta-1}{\zeta}} + (1-\alpha)h_t^{\frac{\zeta-1}{\zeta}}]^{\frac{\zeta}{\zeta-1}} \quad (58)$$

$$i_t = k_{t+1} - (1-\delta)k_t \quad (59)$$

$$ac_t = \frac{\phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t \quad (60)$$

$$(1-\lambda)c_t^{unc} = y_t - i_t - ac_t + d_{t+1}/(1+r^*) - d_t - \lambda h_t^\gamma \quad (61)$$

where we used the fact that $c_t^{htm} = h_t^\gamma$. This allows us to solve for c_t^{unc} , h_t , and k_{t+1} as functions of the aggregate variables. Thereafter, we can back out the variables for each

household type.

The last equation is the recursive equation for the optimization problem:

$$V(a_t, k_t, d_t) = \max_{c_t, h_t, k_{t+1}, d_{t+1}} \left[\frac{(c_t^{unc} - h_t^\gamma / \gamma)^{1-\gamma}}{1-\gamma} + \beta E_t [V(a_{t+1}, k_{t+1}, d_{t+1})] \right] \quad (62)$$

The expectation is taken over next-period productivity conditional on the current productivity state.

We maintain most of the parameters used in the perturbation method. For this global-method exercise, we set $r_t = r^*$ and replace the debt-elastic interest-rate stationarity device with impatience. To induce stationarity, we impose

$$\beta(1 + r^*) < 1 \quad (63)$$

To solve the model using the global solution method, we iterate the value function of the unconstrained households on a grid of productivity, debt, and capital. Following [Uribe and Schmitt-Grohé \(2017\)](#), we use 9 grid points for productivity, 70 grid points for debt, and 30 grid points for capital.

Next, we set the bounds on the capital grid to 0.5 and 1.1. The equilibrium probability distribution of capital assigns positive probability only to capital levels in the range $[0.7069, 0.8931]$, fully contained within the specified bounds.

For debt, we use a grid from 0 to 0.70, matching the range relevant for the baseline calibration. This specification is used to assess whether the central relative-consumption-volatility result is sensitive to the local solution method.

A.5. Complete specification of the simple model with firm dynamics

This section extends the simple model by adding the firm-dynamics block used in Section 3.4. The first part describes the production structure with sectoral entry, varieties, and variable markups; the remaining parts collect the equilibrium conditions and steady state for the augmented economy.

A.5.1. Production side: introduction of firm dynamics in the simple model

The final consumption good, y_t , with a normalized price of $p_t = 1$, is produced by a competitive retailer that aggregates production over a continuum of sectors, j , with

elasticity of substitution $1/(1 - \omega)$:

$$y_t = \left(\int_0^1 y_{jt}^\omega dj \right)^{1/\omega} \quad (64)$$

Production in each sector j is determined by aggregating the output $\{\tilde{y}_{jit}\}$ of n_t intermediate firms, i , operating in a competitive market. The elasticity of substitution across varieties is given by $1/(1 - \tau)$, and sectoral output is defined as follows:

$$y_{jt} = n_t \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \tilde{y}_{jit}^\tau \right)^{1/\tau} \quad (65)$$

Intermediate firms operate in a monopolistically competitive market within each sector. Production requires hiring labor $\tilde{l}_{jit}$, and renting capital $\tilde{\kappa}_{jit}$:

$$\tilde{y}_{jit} = a_t \cdot \left(\alpha \tilde{\kappa}_{jit}^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \tilde{l}_{jit}^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (66)$$

where a_t is a common productivity shock.

To produce this output, firms must set aside φ units of capital as a start-up cost required to initiate production. This capital is not used directly in the production process, but firms must still pay rent on these units. The presence of this fixed capital requirement ensures that there are no economic profits in equilibrium.

Each intermediate producer faces an inverse demand schedule p_{jit} consistent with the optimal decisions made by retailers of the final consumption good (64) and the wholesalers of sectoral-level intermediate goods (65). Profit maximization involves choosing capital and labor to solve:

$$\tilde{\pi}_{jit} = \max_{\tilde{\kappa}_{jit}, \tilde{l}_{jit}} \left\{ p_{jit} \tilde{y}_{jit} - r_{kt}(\tilde{\kappa}_{jit} + \varphi) - w_t \tilde{l}_{jit} \right\} \quad (67)$$

subject to the production technology (66), the demand schedule, and input prices $\{r_{kt}, w_t\}$ that are taken as given. The first-order conditions are summarized as:

$$w_t = p_{jit} \frac{\partial \tilde{y}_{jit}}{\partial \tilde{l}_{jit}} + \tilde{y}_{jit} \frac{\partial p_{jit}}{\partial \tilde{y}_{jit}} \frac{\partial \tilde{y}_{jit}}{\partial \tilde{l}_{jit}} = \frac{p_{jit}}{\mu_{jit}} (1 - \alpha) a_t \left(\tilde{y}_{jit} / (a_t \tilde{l}_{jit}) \right)^{\frac{1}{\zeta}} \quad (68)$$

$$r_{kt} = p_{jit} \frac{\partial \tilde{y}_{jit}}{\partial \tilde{\kappa}_{jit}} + \tilde{y}_{jit} \frac{\partial p_{jit}}{\partial \tilde{y}_{jit}} \frac{\partial \tilde{y}_{jit}}{\partial \tilde{\kappa}_{jit}} = \frac{p_{jit}}{\mu_{jit}} \alpha a_t \left(\tilde{y}_{jit} / \left(a_t \tilde{\kappa}_{jit} \right) \right)^{\frac{1}{\zeta}} \quad (69)$$

where μ_{jit} is the markup for the intermediate producer that depends on the number of firms, n_t , operating in equilibrium:

$$\mu_{jit} = 1 + \frac{(1 - \tau)(1 - \omega)n_t}{\tau(1 - \omega)n_t - (\tau - \omega)} > 1 \quad (70)$$

This formulation demonstrates that the markup decreases as more firms enter the economy (when n_t increases) and increases when firms exit. The variable nature of the markup stems from the finite sectoral structure specified in Equation (65) where an individual producer's decisions y_{ji} influence the sectoral price index p_j but not the general price index $p = 1$. Consequently, when a sector contains only a single firm ($n_t = 1$), the markup equals the sectoral value of $1/\omega$. Conversely, as the number of firms in a sector approaches infinity ($n_t \rightarrow \infty$), the markup converges to the product-level value of $1/\tau$. Since $1/\tau < 1/\omega$, the markup varies endogenously with the equilibrium number of firms, establishing that markups are inherently countercyclical rather than acyclical.

From the production side of the economy, a symmetric equilibrium emerges where the sectoral and total output become:

$$y_t = y_{jt} = n_t \tilde{y}_{jit}, \quad \forall i, j$$

At the same time, due to the normalization of the general price index, intermediate goods prices become $p_{jit} = p_t = 1$. Given the symmetry, we omit the indices j and i in the remainder of this section.

By assumption, firms are free to enter the market, meaning that all profits are driven to zero in equilibrium: $\tilde{\pi}_t = 0$. From the firm profit maximization problem (67) and using the respective equilibrium conditions (68), (69), and (70), we can then determine the size of the industry as:

$$n_t = \frac{a_t(\mu_t - 1)}{\mu_t r_{kt} \varphi} \left(\alpha \kappa_t^{1-1/\zeta} + (1 - \alpha) l_t^{1-1/\zeta} \right)^{\zeta/(\zeta-1)} \quad (71)$$

where $\kappa_t = n_t \tilde{\kappa}_t$ and $l_t = n_t \tilde{l}_t$ capture the aggregate utilization of labor and capital (excluding the fixed cost).

Market clearing requires that households' supply of capital and labor equal input

demand for production:

$$k_t = \kappa_t + n_t \varphi$$

$$h_t = l_t$$

Finally, using the equilibrium conditions and the households' budget constraints (2) and (17), we can derive the overall economy resource constraint as:

$$c_t = y_t - i_t - \frac{\Phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t + \frac{d_{t+1}}{1 + r_t} - d_t$$

A.5.2. Equilibrium conditions

The equilibrium conditions below combine the household blocks from the simple model with the firm-dynamics production structure introduced above. They characterize the augmented economy used to quantify the additional amplification from procyclical firm entry.

Hand-to-mouth households.

$$c_t^{htm} = w_t h_t^{htm} \tag{72}$$

$$w_t = (h_t^{htm})^{\nu-1} \tag{73}$$

Unconstrained households.

$$\left(c_t^{unc} - \frac{(h_t^{unc})^\nu}{\nu} \right)^{-\gamma} = \zeta_t^{unc} \tag{74}$$

$$w_t = (h_t^{unc})^{\nu-1} \tag{75}$$

$$\zeta_t^{unc} = \beta(1 + r_t) E_t[\zeta_{t+1}^{unc}] \tag{76}$$

$$\zeta_t^{unc} \left(1 + \phi \left(\frac{k_{t+1}}{k_t} - 1 \right) \right) = \beta E_t \left[\zeta_{t+1}^{unc} \left\{ 1 - \delta + r_{kt+1} + \frac{\phi}{2} \left(\left(\frac{k f u_{t+1}}{k f u_t} \right)^2 - 1 \right) \right\} \right] \quad (77)$$

Production.

$$\tilde{y}_{jit} = a_t \cdot \left(\alpha \tilde{\kappa}_{jit}^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \tilde{l}_{jit}^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (78)$$

$$r_{kt} = \frac{\alpha a_t}{\mu_t} \left(\frac{\tilde{y}_{jit}}{a_t \tilde{\kappa}_{jit}} \right)^{\frac{1}{\zeta}} \quad (79)$$

$$w_t = \frac{(1 - \alpha) a_t}{\mu_t} \left(\frac{\tilde{y}_{jit}}{a_t \tilde{l}_{jit}} \right)^{\frac{1}{\zeta}} \quad (80)$$

$$\mu_t = 1 + \frac{(1 - \tau)(1 - \omega)n_t}{\tau(1 - \omega)n_t - (\tau - \omega)} \quad (81)$$

$$\tilde{y}_{jit} \frac{\mu_t - 1}{\mu_t} = r_{kt} \varphi \quad (82)$$

Market clearing.

$$k_t = n_t(\tilde{\kappa}_{jit} + \varphi) \quad (83)$$

$$h_t = n_t \tilde{l}_{jit} \quad (84)$$

Aggregation.

$$c_t = \lambda c_t^{htm} + (1 - \lambda) c_t^{unc} \quad (85)$$

Aggregate resource constraint.

$$c_t + k_{t+1} - (1 - \delta)k_t + \frac{\phi}{2} \left(\frac{k_{t+1}}{k_t} - 1 \right)^2 k_t + d_t = \frac{d_{t+1}}{1 + r_t} + n_t \tilde{y}_{jit} \quad (86)$$

Incomes.

$$y_t^{htm} = c_t^{htm} \quad (87)$$

$$Y_t^{unc} = w_t h_t + \frac{r_{kt} k_t}{1 - \lambda} \quad (88)$$

Closing the model.

$$\log a_{t+1} = \rho_a \log a_t + \sigma_a \epsilon_{at+1} \quad (89)$$

$$r_t = r^* + \psi(e^{\frac{d_{t+1} - \bar{d}}{y}} - 1) \quad (90)$$

Other variable definitions.

$$k f u_t = k_{t+1} \quad (91)$$

$$y_t = w_t h_t + r_{kt} k_t \quad (92)$$

$$I_t = k_{t+1} - (1 - \delta) k_t \quad (93)$$

$$sh_t^h = \frac{w_t h_t}{y_t} \quad (94)$$

$$tby_t = \frac{d_t - \frac{d_{t+1}}{1+r_t}}{y_t} \quad (95)$$

A.5.3. Steady state

In the steady state with no shock ($\epsilon_{at} = 0, \forall t$)

$$a_{ss} = 1$$

$$d_{ss} = \bar{d}$$

$$r_{ss} = r^*$$

$$\beta(1 + r_{ss}) = 1$$

$$r_{kss} = \delta + r_{ss}$$

The values of four steady-state variables: $\tilde{l}_{ss}$, $\tilde{\kappa}_{ss}$, $\tilde{y}_{ss}$, and μ_{ss} can be obtained from the following four equations (given the value of r_{kss}):¹⁴

$$\frac{\alpha}{\mu_{ss}} \left(\frac{\tilde{y}_{ss}}{\tilde{\kappa}_{ss}} \right)^{\frac{1}{\zeta}} = r_{kss}$$

$$\frac{1 - \alpha}{\mu_{ss}} \left(\frac{\tilde{y}_{ss}}{\tilde{l}_{ss}} \right)^{\frac{1}{\zeta}} = (n_{ss} \tilde{l}_{ss})^{\nu-1}$$

$$\tilde{y}_{ss} = \left(\alpha \tilde{\kappa}_{ss}^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) \tilde{l}_{ss}^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}}$$

$$\tilde{y}_{ss} \frac{\mu_{ss} - 1}{\mu_{ss}} = r_{kss} \varphi$$

Given the values of these four variables, the remaining steady-state variables are given as:

$$n_{ss} = \frac{(\tau - \omega) r_{kss} \varphi}{(1 - \omega)(r_{kss} \varphi - \tilde{y}_{ss}(1 - \tau))}$$

$$h_{ss} = n_{ss} \tilde{l}_{ss}$$

$$w_{ss} = h_{ss}^{\nu-1}$$

$$k_{ss} = n_{ss}(\tilde{\kappa}_{ss} + \varphi)$$

$$c_{ss}^{htm} = w_{ss} h_{ss}$$

$$y_{ss}^{htm} = c_{ss}^{htm}$$

$$y_{ss}^{unc} = w_{ss} h_{ss} + \frac{r_{kss} k_{ss}}{1 - \lambda}$$

¹⁴We also use the zero profit condition to get n_{ss} for the second equation: $n_{ss} = \frac{(\tau - \omega) r_{kss} \varphi}{(1 - \omega)(r_{kss} \varphi - \tilde{y}_{ss}(1 - \tau))}$ and $\mu_{ss} = \frac{\tilde{y}_{ss}}{\tilde{y}_{ss} - r_{kss} \varphi}$

$$y_{ss} = w_{ss}h_{ss} + r_{kss}k_{ss}$$

$$c_{ss} = y_{ss} - \delta k_{ss} - \frac{r_{ss}\bar{d}}{1 + r_{ss}}$$

$$c_{ss}^{unc} = \frac{c_{ss} - \lambda c_{ss}^{htm}}{1 - \lambda}$$

$$kfu_{kss} = k_{ss}$$

$$I_{ss} = \delta k_{ss}$$

$$sh_{ss}^h = \frac{w_{ss}h_{ss}}{y_{ss}}$$

$$tby_{ss} = \frac{r_{ss}\bar{d}}{(1 + r_{ss})y_{ss}}$$

$$\zeta_{ss}^{unc} = \left(c_{ss}^{unc} - \frac{(h_{ss})^\nu}{\nu} \right)^{-\gamma}$$

$$\zeta_{ss}^{htm} = \left(c_{ss}^{htm} - \frac{(h_{ss})^\nu}{\nu} \right)^{-\gamma}$$

A.6. Complete specification of the full model

This section presents the full estimated model used in Section 4. Relative to the simple model with firm dynamics, the full model adds stochastic trend growth, an interest-rate shock, and a working capital requirement. We first describe these additions and then list the stationary equilibrium conditions and steady state used in estimation.

A.6.1. Additions to the simple model with firm dynamics

The full model preserves the firm-dynamics structure of Appendix A.5 and adds the stochastic components needed for the Bayesian estimation. This subsection introduces each additional block before the full stationary equilibrium system is stated below.

Trend shock. In addition to the transitory shock to output, the full model incorporates an exogenous stochastic trend growth variable, denoted as x_t , which evolves according to the realization of the stochastic process $\{g_t\}$:

$$g_{t+1} = x_{t+1}/x_t$$

where the logarithm of g_t follows an AR(1) process:

$$\ln g_{t+1} = (1 - \rho_g) \ln g + \rho_g \ln g_t + \sigma_g \varepsilon_{gt+1}, \quad \varepsilon_{gt+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

for a steady-state trend growth rate $g > 0$.

The trend variable enters household preferences:

$$u(c_t^i, h_t^i) = (c_t^i - x_{t-1}(h_t^i)^\nu/\nu)^{1-\gamma} / (1-\gamma) \quad (96)$$

and the production function:

$$\tilde{y}_{jit} = a_t \cdot \left(\alpha \tilde{\kappa}_{jit}^{\frac{\zeta-1}{\zeta}} + (1-\alpha) (x_t \tilde{l}_{jit})^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (97)$$

The inclusion of the trend growth variable in both preferences and production ensures that the model exhibits balanced growth, where output, consumption, and physical capital grow at the same rate while hours worked remain constant in the steady state.

Interest rate shock and working capital constraint. The third shock introduced into the model, $\{\xi_t\}$, follows an AR(1) process:

$$\log \xi_{t+1} = \rho_\xi \log \xi_t + \sigma_\xi \varepsilon_{\xi t+1}, \quad \varepsilon_{\xi t+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

This shock introduces stochastic fluctuations in the external interest rate, which is specified as:

$$r_t = r^* + \psi \cdot \left(e^{\frac{d_{t+1}/x_t - \bar{d}}{\bar{y}}} - 1 \right) + e^{\xi_t - 1} - 1$$

A further friction is introduced by assuming that firms must finance a fraction η of their wage bill through external borrowing. This working capital constraint requires firms to obtain loans of $\eta w_t \tilde{l}_{jit} / (1 + r_t)$ at the beginning of the period, which they repay at

the end of the period. The associated interest expense, $w_t \tilde{l}_{jit} \eta r_t / (1 + r_t)$, increases the effective cost of labor, thereby reducing labor demand. Consequently, the firm's optimal labor choice differs from that in equation (68).

$$w_t \cdot (1 + \eta r_t / (1 + r_t)) = \frac{(1 - \alpha) x_t a_t}{\mu_t} \left(\tilde{y}_{jit} / (x_t a_t \tilde{l}_{jit}) \right)^{\frac{1}{\zeta}}$$

Given the presence of trend growth, the model is inherently nonstationary. To obtain a well-defined deterministic steady state and ensure stationary rational expectations dynamics, we transform all nonstationary variables at time t by normalizing them with respect to x_{t-1} .

Denoting transformed variables with a hat notation, the trend variable itself is redefined as $\hat{x}_t = x_t / x_{t-1} = g_t$, and other key variables are adjusted accordingly: $\hat{v}_t = v_t / x_{t-1}$.

A.6.2. Equilibrium conditions

The equilibrium conditions are written in stationary form after normalizing the growing variables by the trend component. The system below combines the household conditions, production-side optimality conditions, market clearing, and exogenous shock processes that enter the likelihood.

Hand-to-mouth households.

$$\hat{c}_t^{htm} = \hat{w}_t h_t^{htm} \quad (98)$$

$$\hat{w}_t = (h_t^{htm})^{\nu-1} \quad (99)$$

Unconstrained households.

$$\left(\hat{c}_t^{unc} - \frac{(h_t^{unc})^\nu}{\nu} \right)^{-\gamma} = \hat{\zeta}_t^{unc} \quad (100)$$

$$\hat{w}_t = (h_t^{unc})^{\nu-1} \quad (101)$$

$$g_t^\gamma \hat{\zeta}_t^{unc} = \beta(1 + r_t) E_t[\hat{\zeta}_{t+1}^{unc}] \quad (102)$$

$$g_t^\gamma \hat{\zeta}_t^{unc} \left(1 + \phi \left(\frac{g_t \hat{k}_{t+1}}{\hat{k}_t} - g \right) \right) = \beta E_t \left[\hat{\zeta}_{t+1}^{unc} \left\{ 1 - \delta + r_{kt+1} + \frac{\phi}{2} \left(\left(\frac{g_t \hat{k} \hat{f} u_{t+1}}{\hat{k} \hat{f} u_t} \right)^2 - g^2 \right) \right\} \right] \quad (103)$$

Production.

$$\hat{y}_{jit} = a_t \left(\alpha \hat{\kappa}_{jit}^{\frac{\zeta-1}{\zeta}} + (1 - \alpha) (g_t \tilde{l}_{jit})^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}} \quad (104)$$

$$r_{kt} = \frac{\alpha a_t}{\mu_t} \left(\frac{\hat{y}_{jit}}{a_t \hat{\kappa}_{jit}} \right)^{\frac{1}{\zeta}} \quad (105)$$

$$\hat{w}_t \left(1 + \frac{\eta r_t}{1 + r_t} \right) = \frac{(1 - \alpha) g_t a_t}{\mu_t} \left(\frac{\hat{y}_{jit}}{a_t g_t \tilde{l}_{jit}} \right)^{\frac{1}{\zeta}} \quad (106)$$

$$\mu_t = \frac{(1 - \omega) n_t - (\tau - \omega)}{\tau(1 - \omega) n_t - (\tau - \omega)} \quad (107)$$

$$\hat{y}_{jit} \frac{\mu_t - 1}{\mu_t} = r_{kt} \hat{\varphi} \quad (108)$$

Market clearing.

$$\hat{k}_t = n_t (\hat{\kappa}_{jit} + \hat{\varphi}_t) \quad (109)$$

$$h_t = n_t \tilde{l}_{jit} \quad (110)$$

Aggregation.

$$\hat{c}_t = \lambda \hat{c}_t^{htm} + (1 - \lambda) \hat{c}_t^{unc} \quad (111)$$

Aggregate resource constraint.

$$\hat{c}_t + g_t \hat{k}_{t+1} - (1 - \delta) \hat{k}_t + \frac{\phi}{2} \left(\frac{g_t \hat{k}_{t+1}}{\hat{k}_t} - g \right)^2 \hat{k}_t + \hat{d}_t = \frac{g_t \hat{d}_{t+1}}{1 + r_t} + n_t \hat{y}_{jit} - \frac{\eta r_t}{1 + r_t} \hat{w}_t h_t \quad (112)$$

Incomes.

$$\hat{y}_t^{htm} = \hat{c}_t^{htm} \quad (113)$$

$$\hat{y}_t^{unc} = \hat{w}_t h_t + \frac{r_{kt} \hat{k}_t}{1 - \lambda} \quad (114)$$

Closing the model.

$$\log a_{t+1} = \rho_a \log a_t + \sigma_a \epsilon_{at+1} \quad (115)$$

$$\ln(g_{t+1}/g) = \rho_g \ln(g_t/g) + \sigma_g \epsilon_{gt+1} \quad (116)$$

$$\log \xi_{t+1} = \rho_\xi \log \xi_t + \sigma_\xi \epsilon_{\xi t+1} \quad (117)$$

$$r_t = r^* + \psi \left(e^{\frac{\hat{d}_{t+1} - \bar{d}}{\bar{y}}} - 1 \right) + (e^{\xi_t - 1} - 1) \quad (118)$$

Other variable definitions.

$$k \hat{f} u_t = g_t \hat{k}_{t+1} \quad (119)$$

$$\hat{y}_t = n_t \hat{y}_{jit} - \frac{\eta r_t}{1 + r_t} \hat{w}_t h_t \quad (120)$$

$$\hat{I}_t = g_t \hat{k}_{t+1} - (1 - \delta) \hat{k}_t \quad (121)$$

$$sh_t^h = \frac{\hat{w}_t h_t}{\hat{y}_t} \quad (122)$$

$$tby_t = \frac{\hat{d}_t - \frac{g_t \hat{d}_{t+1}}{1 + r_t}}{\hat{y}_t} \quad (123)$$

A.6.3. Steady state

In the steady state with no shock ($\epsilon_{it} = 0$, $\forall t, i \in \{a, g, \xi\}$)

$$a_{ss} = 1$$

$$g_{ss} = g$$

$$\kappa_{ss} = 1$$

$$r_{ss} = r^*$$

$$\beta(1 + r^*) = g^\gamma$$

$$r_{kss} = \delta + r_{ss}$$

The values of four steady-state variables: $\tilde{l}_{ss}$, $\hat{\kappa}_{ss}$, $\hat{y}_{ss}$, and μ_{ss} can be obtained from the following four equations (given the value of r_{kss}):¹⁵

$$\frac{\alpha}{\mu_{ss}} \left(\frac{\hat{y}_{ss}}{\hat{\kappa}_{ss}} \right)^{\frac{1}{\zeta}} = r_{kss}$$

$$\frac{1-\alpha}{\mu_{ss}} g_{ss} \left(\frac{\hat{y}_{ss}}{g_{ss} \tilde{l}_{ss}} \right)^{\frac{1}{\zeta}} = (n_{ss} \tilde{l}_{ss})^{\gamma-1} \left(1 + \frac{\eta r_{ss}}{1 + r_{ss}} \right)$$

$$\hat{y}_{ss} = \left(\alpha \hat{\kappa}_{ss}^{\frac{\zeta-1}{\zeta}} + (1-\alpha)(g_{ss} \tilde{l}_{ss})^{\frac{\zeta-1}{\zeta}} \right)^{\frac{\zeta}{\zeta-1}}$$

$$\hat{y}_{ss} \frac{\mu_{ss} - 1}{\mu_{ss}} = r_{kss} \hat{\phi}$$

Given the values of these four variables, the remaining steady-state variables are given as:

$$n_{ss} = \frac{(\tau - \omega) r_{kss} \hat{\phi}}{(1 - \omega)(r_{kss} \hat{\phi} - \hat{y}_{ss}(1 - \tau))}$$

$$h_{ss} = n_{ss} \tilde{l}_{ss}$$

$$\hat{w}_{ss} = h_{ss}^{\gamma-1}$$

¹⁵We also use the zero profit condition to get n_{ss} for the second equation: $n_{ss} = \frac{(\tau - \omega) r_{kss} \hat{\phi}}{(1 - \omega)(r_{kss} \hat{\phi} - \hat{y}_{ss}(1 - \tau))}$ and

$$\mu_{ss} = \frac{\hat{y}_{ss}}{\hat{y}_{ss} - r_{kss} \hat{\phi}}$$

$$\hat{k}_{ss} = n_{ss}(\hat{\kappa}_{ss} + \hat{\varphi})$$

$$\hat{c}_{ss}^{htm} = \hat{w}_{ss}h_{ss}$$

$$\hat{y}_{ss}^{htm} = \hat{c}_{ss}^{htm}$$

$$\hat{y}_{ss}^{unc} = \hat{w}_{ss}h_{ss} + \frac{r_{kss}\hat{k}_{ss}}{1 - \lambda}$$

$$\hat{y}_{ss} = n_{ss}\hat{y}_{ss} - \frac{\eta r_{ss}}{1 + r_{ss}}\hat{w}_{ss}h_{ss}$$

$$\hat{c}_{ss} = \hat{y}_{ss} - (g_{ss} - 1 + \delta)\hat{k}_{ss} - \frac{(1 + r_{ss} - g_{ss})}{1 + r_{ss}}\bar{d}$$

$$\hat{c}_{ss}^{unc} = \frac{\hat{c}_{ss} - \lambda\hat{c}_{ss}^{htm}}{1 - \lambda}$$

$$kf u_{ss} = g_{ss}\hat{K}_{ss}$$

$$\hat{I}_{ss} = (g_{ss} - 1 + \delta)\hat{k}_{ss}$$

$$sh_{ss}^h = \frac{\hat{w}_{ss}h_{ss}}{\hat{y}_{ss}}$$

$$tby_{ss} = \frac{(1 + r_{ss} - g_{ss})}{1 + r_{ss}} \frac{\bar{d}}{\hat{y}_{ss}}$$

$$\hat{\zeta}_{ss}^{unc} = \left(\hat{c}_{ss}^{unc} - \frac{(h_{ss})^\nu}{\nu} \right)^{-\gamma}$$

A.7. Estimation results: posterior distributions

This section reports the country-level posterior means from the Bayesian estimation of the full model. Table 10 lists the prior ranges and posterior means for the structural

parameters and the measurement-error terms used in the estimation.

Table 10: Estimated parameters of the full model

Parameter	σ_a	ρ_a	σ_g	ρ_g	σ_ξ	ρ_ξ	ϕ	ψ	η	ω	τ	λ
Prior distribution												
Min.	0.0000	-0.9900	0.0000	-0.9900	0.0000	-0.9900	0.0000	0.0000	0.0000	0.0000	0.5000	0.0000
Max.	0.2000	0.9900	0.2000	0.9900	0.2000	0.9900	5.0000	2.0000	1.0000	0.5000	1.0000	0.9900
Posterior distribution: means												
Low-income countries												
Benin	0.0161	0.8818	0.0142	0.9858	0.1329	0.8376	2.3304	0.4045	0.1487	0.2696	0.9523	0.4564
Burkina Faso	0.0128	0.9085	0.0116	0.9842	0.1014	0.7507	2.4861	0.1169	0.2225	0.3280	0.8259	0.6087
Burundi	0.0274	0.8746	0.0250	0.9864	0.1409	0.8979	2.8039	0.2017	0.1075	0.2506	0.9926	0.8809
Cameroon	0.0318	0.7128	0.0425	0.9879	0.0532	0.9451	2.2913	0.8613	0.5620	0.2648	0.8939	0.8236
Central African Republic	0.0299	0.7141	0.0335	0.9848	0.0638	0.6184	1.2104	0.3924	0.3148	0.2510	0.9584	0.7690
India	0.0113	0.5976	0.0161	0.9776	0.0524	0.8572	1.3818	1.2812	0.3714	0.2249	0.9936	0.7223
Indonesia	0.0140	0.8623	0.0080	0.9872	0.0228	0.6983	0.6017	0.0012	0.1101	0.1131	0.9839	0.0482
Kenya	0.0167	0.4494	0.0263	0.9882	0.0702	0.9643	2.8348	0.3923	0.3550	0.2817	0.8832	0.8354
Niger	0.0264	0.7069	0.0188	0.9795	0.0532	0.8799	0.6913	0.2230	0.4176	0.2351	0.9941	0.5049
Pakistan	0.0085	0.8246	0.0053	0.9823	0.0320	0.9139	3.0047	0.0269	0.1740	0.2646	0.9258	0.4075
Senegal	0.0136	0.5414	0.0058	0.9842	0.0255	0.8526	1.2507	0.0584	0.2016	0.2539	0.9805	0.3945
Emerging countries												
Algeria	0.0335	0.8557	0.0400	0.9884	0.0493	0.9134	3.7118	0.4639	0.2349	0.4085	0.7262	0.1322
Botswana	0.0217	0.9233	0.0039	0.9873	0.0289	0.8730	1.5793	0.0004	0.1106	0.2697	0.8906	0.1619
Chile	0.0211	0.8545	0.0096	0.9846	0.0390	0.1889	0.7716	0.0029	0.3311	0.2163	0.9828	0.0593
Costa Rica	0.0161	0.8469	0.0201	0.9884	0.0809	0.7484	3.2838	0.5473	0.1662	0.3288	0.8428	0.8132
Cuba	0.0459	0.8626	0.0574	0.9417	0.0743	0.8461	0.9546	1.5318	0.6593	0.2475	0.9689	0.8615
Egypt, Arab Rep	0.0090	0.7145	0.0031	0.9883	0.0096	0.9503	0.3715	0.0010	0.0791	0.4515	0.9450	0.6422
El Salvador	0.0194	0.9031	0.0186	0.9861	0.0589	0.8884	2.5681	0.2356	0.1459	0.2829	0.9440	0.8354
Iran, Islamic Rep	0.0526	0.7413	0.0531	0.9868	0.0671	0.8804	1.3155	0.6869	0.2491	0.4345	0.7664	0.5767
Korea, Rep	0.0394	0.5706	0.0414	0.5216	0.0137	0.4992	0.0751	0.2252	0.4838	0.0945	0.6424	0.6545
Malaysia	0.0144	0.9236	0.0033	0.9850	0.0209	0.8144	1.6959	0.0001	0.2427	0.3587	0.8780	0.1637
Mexico	0.0109	0.8118	0.0122	0.9860	0.0100	0.8959	0.9495	0.1366	0.8275	0.1653	0.9285	0.2268
Morocco	0.0197	0.7944	0.0156	0.9832	0.0673	0.8764	2.6017	0.4057	0.2210	0.4131	0.7363	0.8085
Panama	0.0209	0.8756	0.0230	0.9873	0.1292	0.8713	2.6105	0.2553	0.3118	0.3121	0.9001	0.7884
South Africa	0.0120	0.7442	0.0141	0.9859	0.0346	0.9524	0.8225	1.2535	0.6509	0.3424	0.8225	0.0677
Thailand	0.0119	0.8608	0.0063	0.9809	0.0231	0.7591	1.5245	0.0108	0.4631	0.4456	0.8657	0.2180
Tunisia	0.0248	0.8421	0.0226	0.9875	0.0501	0.9281	4.1423	0.3204	0.2147	0.3663	0.8047	0.8199
Turkey	0.0157	0.8346	0.0214	0.9862	0.0227	0.8386	1.2176	0.2633	0.6900	0.3718	0.9145	0.7643
Advanced countries												
Austria	0.0071	0.8158	0.0018	0.9828	0.0043	0.7460	0.6968	0.0029	0.5091	0.3631	0.9347	0.0630
Belgium	0.0089	0.8781	0.0014	0.9827	0.0065	0.7949	0.7830	0.0014	0.1435	0.3715	0.9734	0.2303
Denmark	0.0091	0.8955	0.0083	0.9559	0.0180	0.7819	0.2052	1.5387	0.2881	0.2195	0.9804	0.2027
Finland	0.0146	0.8729	0.0021	0.9798	0.0119	0.7065	1.0474	0.0007	0.1587	0.2800	0.9124	0.1103
Germany	0.0085	0.8076	0.0013	0.9836	0.0047	0.7535	0.7713	0.0010	0.6354	0.2971	0.9753	0.1128
Hong Kong SAR, China	0.0192	0.9173	0.0025	0.9862	0.0212	0.7949	1.5151	0.0000	0.1067	0.3091	0.9146	0.1963
Italy	0.0108	0.8112	0.0087	0.9795	0.0202	0.9021	1.4479	0.7962	0.4738	0.3002	0.9295	0.4633
Netherlands	0.0140	0.9856	0.0027	0.9818	0.0491	0.3303	1.0519	0.3994	0.9228	0.2558	0.9648	0.0525
Sweden	0.0108	0.7978	0.0028	0.9832	0.0107	0.6322	0.9001	0.0025	0.1717	0.2780	0.9427	0.0732
United Kingdom	0.0100	0.8052	0.0094	0.9628	0.0122	0.4695	0.4115	1.3708	0.6970	0.2375	0.9685	0.0962